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Unraveling the effects of micro-level street environment on dockless bikeshare in Ithaca

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A R T I C L E I N F O

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A B S T R A C T

Trip trajectory data from dockless bikeshare (DBS) users have enabled new avenues to investigate the impacts of built environments on cycling behaviors. Nevertheless, most prior studies focused on macro-level attributes (e.g., land use), overlooking the micro-level streetscape characteristics that influence cyclists' in-situ perceptions. Although studies have explored the role of objective visual features (e.g., greenery), few have addressed that of perceived visual qualities (e.g., enclosure). Using ∼ 5,000 Lime trips, we found both objective visual features and subjective perceptions significantly affect DBS traffic in understudied small cities like Ithaca. Wider and open streets with higher visual complexity, human scale, and safety attract DBS users, consistent with docked bikeshare and personal bike findings. Conversely, streets characterized by higher enclosure, and excessive obstructions (e.g., cars and grass) deter cyclists. Our study provides evidence for planners to renovate bicycle-friendly streets with minimal urban design interventions to foster more active travel.

1. Introduction

1.1. Background

Bikeshare has gained wide popularity by bringing multifaceted benefits as an affordable, sustainable, and active transport mode (Chen et al., 2020b; Shaheen et al., 2010; Shen et al., 2018). More recently, the emerging dockless bikeshare (DBS), an alternative to the traditional docked model, has revolutionized the sector (Gu et al., 2019; Guo et al., 2022). Its inherent flexibility regarding parking and adaptable fleet sizes has garnered a broader appeal, signifying an immense potential for promoting active travel (Gu et al., 2019; Shen et al., 2018). Consequently, the DBS has captivated considerable academic interests in the transportation and urban planning literature, with remarkable efforts paid to uncover the complexity of travel behaviors, the interplay with public transit (Chen and Ye, 2021; Gao et al., 2023), and the relationship with the built environment (BE) (Qiu and Chang, 2021; Wei et al., 2023).

Meanwhile, the integration of GPS devices into the DBS system has significantly automated the collection of route data, offering

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opportunities to delve into how the BE affects cycling patterns with finer granularity and in a sizeable manner (Guo and He, 2021; Koh and Wong, 2013; Lue and Miller, 2019; Sevtsuk et al., 2021). However, existing research predominantly deploys macro-level BE attributes (e.g., land use) (Faghih-Imani et al., 2014; Faghih-Imani and Eluru, 2016; Noland et al., 2016; Qiu and Chang, 2021), while overlooking the intricacies of the micro-level street environment (SE) characteristics. This oversight frames a critical gap, as it is essential to understand cyclists' in-situ perceptions at the micro-level SE (Blitz, 2021; Ewing and Handy, 2009; Guo and He, 2021), where perceived environment characteristics play a direct role in mediating the impact of the objective physical environment on cycling behavior (Ma et al., 2014; Ma and Dill, 2015). Such invaluable insights on perceived micro-level SE will formulate evidence for urban planners and designers to envision more desirable streets through cost-effective streetscape design interventions to change perceptions directly (Mertens et al., 2016; Shen et al., 2018). Although several attempts have been made to examine how micro-level SE attributes influence pedestrian and runner route choices (Dong et al., 2023; Salazar Miranda et al., 2021; Sevtsuk et al., 2021), the impact on cyclists remains largely underexplored (Wang et al., 2020; Wei et al., 2023). We speculate that perceptions of walkers, runners, and bikers differ vastly over the exact same street scene due to the varying trip characteristics regarding speed and purpose. Specifically, we identified three critical gaps concerning the influence of micro-level SE attributes on DBS cycling.

1.2. Research gaps

First, the impact of subjectively measured perceptions on street scenes along biking routes needs to be addressed. With emerging big data such as Street View Imagery (SVI) and its integration with Machine Learning (ML) algorithms, opportunities arise to accurately depict micro-level visual features related to cycling trips (Ito and Biljecki, 2021). However, individual street features proxied through oversimplified view indices are unsatisfactory because they fail to convey the subjective experience of a cyclist when she/he immerses in and interacts with the immediate street environment. Researchers argued that one separate feature showing a positive impact may be offset by other elements that pose adverse effects in the exact same scenes (Zhao et al., 2022). After all, human perception is subjective – a sensory process. In that regard, the citywide perception data, particularly perceived urban design qualities (e.g., enclosure) collected through visual surveys, can offer stronger predictive powers than individual visual elements in explaining various human behaviors vital to multifaceted urban aspects, including walkability, housing prices, and vitality (Ewing and Handy, 2009; Qiu et al., 2022; Song et al., 2022b; Wang et al., 2023). The complexity of human perceptions renders it imperative to consider both subjective perceived qualities and objective physical elements to accurately describe micro-level SE attributes (Ramírez et al., 2021). Furthermore, while streets are where pedestrians, runners, and cyclists directly share and immerse themselves (Porter et al., 2018; Van Dyck et al., 2012), DBS biking is considerably the fastest mode for commuting as the primary purpose, raising questions about its divergent roles and strength of associations with micro-level SE compared to walkers and runners. Despite the importance of subjective perceptions, there remains a paucity of empirical evidence on how these qualities relate to DBS traffic at the micro-scale (Koh and Wong, 2013; Song et al., 2023b).

Second, ample knowledge has been accrued on the impacts of macro-level BE (e.g., land use, POI, street connectivity) on affecting cycling behavior (Faghih-Imani and Eluru, 2015; Lin et al., 2018; Noland et al., 2016; Qiu and Chang, 2021), while the role of different micro-level SE visual features remains neglected. Previous studies have utilized computer vision (CV) techniques to semantically extract visual elements from SVIs and identified the impacts of limited ubiquitous features on cycling, such as sky view and green view (Lu et al., 2019; Zhang et al., 2023). However, other essential visual features, like street furniture, are yet to receive adequate attention (Wei et al., 2023). Meanwhile, although pedestrian behavior studies (Salazar Miranda et al., 2021) have indicated the non-negligible role of streetlights, street furniture, and subtle features like fountains, their impacts on cycling may be minor in light of pedestrians' slower speeds. There is still uncertainty, however, as to whether and to what extent subtle streetscape visual features, such as furniture and plants, affect DBS cycling at the micro-level. Such oversight warrants a systematic and comprehensive investigation of micro-level SE visual features, encompassing both ubiquitous (e.g., sky, building) and subtle ones (e.g., streetlight, grass, furniture) on cycling behaviors.

Third, the North American context has been largely missing in prior studies that have utilized GPS trajectories to investigate DBS cycling behaviors, mainly owing to the DBS' prevalence in Asia, despite its increasing share of the bikeshare market (Bike Share in the U.S., 2017). Moreover, studies on cycling behaviors have predominantly examined large cities like Beijing, Singapore, and Montreal (Faghih-Imani et al., 2014; Shen et al., 2018; Zhao and Li, 2017), ignoring behaviors in small-sized cities (Qiu and Chang, 2021). Our study posits that urban morphology, which varies considerably between large cities and small towns, can have a discrete impact on cycling behaviors, sometimes even in opposite ways. For instance, land use mix is a stronger predictor of bikeshare use around workplaces in dense urban areas than in low-density cities (Zhao and Li, 2017). Hence, empirical evidence from small North American cities is necessary to enrich our knowledge of the influence of micro-level SE on DBS cycling.

1.3. Research objectives

The study aims to advance our understanding of how micro-level SE characteristics influence cycling volume at the street segment level. Particularly, when controlling the effects of macro-level BE, it explores the roles of visual features and perceived qualities of micro-level SE. Furthermore, it meticulously compares the explanatory power and directions of these characteristics across a range of urban contexts and modes of active travel. It offers meaningful insights into the physical elements and urban design qualities that can facilitate desirable and biking-friendly streets, contributing significantly to the broader literature on DBS and active transportation for an enhanced quality of life.

To accomplish these ambitious objectives, this study leverages a large DBS trajectory dataset in Ithaca (Qiu and Chang, 2020), a small college town in upstate New York. We seek to answer the following research questions explicitly: 1) To what extent does the human perception of micro-level SE complement macro-level BE attributes in explaining DBS trip volume? 2) Do micro-level SE visual features and perceived qualities conflict with or complement each other in predicting DBS traffic? 3) How do the explanatory power and signs of micro-level SE attributes in small cities converge with or diverge from those of DBS users in large metropolitan areas, other active travel modes (i.e., walking, running), and other cycling service types (i.e., docked systems, personal bikes)?

The contribution of this research is four-fold, which includes: 1) For the first time, revealing the role of micro-level SE perceptions, with a primary focus on urban design qualities, on DBS traffic volume at the street segment scale using a scalable ML method based on SVIs. The results provide valuable practical implications for facilitating desirable and biking-friendly streets. 2) Enriching the DBS literature by comprehensively identifying the effect of micro-level SE visual features, encompassing both ubiquitous and poorly addressed subtle ones. 3) Extending the findings to compare the impacts of micro-level SE on DBS with other modes like running, offering a replicable analytical framework for future research to study micro-level SE attributes and active transportation behaviors 4) Leveraging a large GPS cycling trajectory dataset to investigate DBS cyclists' route preferences in small-city contexts.

2. Literature review

2.1. Perceived street environment and cycling behaviors

According to environmental psychology theory (Mehrabian and Russell, 1974), individual perceptions of the environment could influence emotions, subsequently impacting behavioral decisions. This idea is echoed in recent public health and transportation research, underscoring the mediating role of perceptions between the physical street environment and travel behavior (Ettema, 2016; Koh and Wong, 2013; Ma et al., 2014; Panter and Ogilvie, 2015). Given this relationship, a growing body of literature has been conducted to investigate the connection between perceived BE characteristics and biking behaviors at a macro-level (Blitz, 2021; Guo and He, 2021, 2020; Mertens et al., 2016; Porter et al., 2018; Van Dyck et al., 2012) For instance, Porter et al. (2018) found strong correlations between perceived residential density, destinations, and street connectivity and transport-related cycling. Similarly, Guo & He (2021) disclosed positive correlations between perceptions of accessibility (to metro stations), availability (of facilities), supportability (bikeways along DBS-metro transfer trips), and cycling; conversely, perceived intersections were negatively correlated. In addition, other attributes, including perceived land use diversity and proximity to places for shopping and recreation, were also reported to be associated with cycling behaviors (Blitz, 2021; Van Dyck et al., 2012).

Although prior studies have scrutinized the effects of perceived BE at different scales, on the one hand, a limited number of micro-level street environment visual features have been considered, including the presence and condition of cycling paths, dedicated streetside parking spaces, and perceived dirt (Blitz, 2021; Guo and He, 2021; Koh and Wong, 2013; Mertens et al., 2016). Nevertheless, how other essential streetscape visual elements, such as furniture and building texture, could affect cycling route choices remains to be addressed. On the other hand, very little is known about the effects of perceived subjective qualities on cycling behaviors, with limited attention paid to the quality or pleasantness of the neighborhood, safety, ease of cycling, and perceived low traffic speed (Blitz, 2021; Guo and He, 2021; Ma et al., 2014; Mertens et al., 2016). Such scarcity on both visual features and subjective qualities can be attributed to the constraints of obtaining the micro-level SE dataset. Previous perception studies primarily relied on self-reported surveys, interviews, and field observations, which are laborious and time-consuming, inevitably confining the studies to small scales to achieve reasonable trade-offs between levels of detail and scalability (Ramírez et al., 2021), rendering it challenging to generalize to a larger context (Cui et al., 2023; Qiu et al., 2022).

2.2. Opportunities of using SVI and CV

Over the past decade, the availability of SVIs and advancements in CV models have opened new avenues for addressing the challenges associated with understanding the micro-level SE to predict human perceptions across various urban regions from a human-centered perspective (Ito and Biljecki, 2021; Zhang et al., 2018). Notably, emerging travel behavior (e.g., walking, running, cycling) studies have extracted view indices (i.e., pixel ratios) of greenery, sky, and road using CV algorithms to describe the micro-level street features present in the scenes quantitatively (Dong et al., 2023; Lu et al., 2019; Sevtsuk et al., 2021; Wang et al., 2022; Wu et al., 2023). Furthermore, Lu et al. (2019) demonstrated a positive correlation between the likelihood of choosing cycling and visual greenness. Zhang et al. (2023) observed that the visual proportions of sky, trees, and pavements in SVIs are positively associated with longer trips and higher speeds. Besides reporting the congruent influence of greenery, Wei et al. (2023) expanded the variables, unraveling that while sidewalks do not significantly affect cycling speed or trip density, streetlight presence might deter cycling during the day. These findings collectively highlight the complex relationship between cycling behaviors and the nuances of micro-level streetscapes measured using SVIs.

Nevertheless, the studies aforementioned only relied on the objective measure of streetscapes, neglecting the role of users' subjective perceptions in situ, which can exhibit stronger strengths of associations with behaviors (Song et al., 2023b). We hypothesize that perceptions of the micro-level SE result from the subtle interplay between an individual's past experience, cultural background, and the physical setting/composition of the street scenes (which can be objectively measured) (Ewing and Handy, 2009). Such a subtle interaction is indeed a complex process of understanding the human brain and psychological sensory information – an inherently subjective concept (Ramírez et al., 2021).

In that regard, the view index of a visual feature – e.g., a tree (Ito and Biljecki, 2021) or the simple recombination of multiple features (Ma et al., 2021) cannot correctly represent the subtle interactions between users and the environment (Ewing and Handy, 2009; Song et al., 2023a). For example, the structures and proportions of aggregated visual components have subtle relationships to human perception of street qualities. An excellent example is the ratio of building heights to total street widths. When it ranges from 3:2 to as low as 1:6, the street scene can impart a comfortable sense of enclosure, a perception concept that has been turned into subjective measures. Moreover, urban designers have defined perceived qualities such as human scale, imageability, and complexity, among others, to be integrated by visual surveys using crowdsourced web platforms using SVI data and artificial intelligence models, including CV and ML (Ewing and Handy, 2009; Qiu et al., 2022). Specifically, Qiu et al. (2023) collected and predicted perceived urban design qualities across Shanghai through visual surveys and ML, revealing their nonnegligible effects on housing prices. Moreover, perceived design qualities not only affect residents' emotions (Zhao et al., 2024), but also influence various human behaviors on urban streets, including walking and running, and their effects might significantly differ from individual visual elements (Dong et al., 2023; Salazar Miranda et al., 2021). A recent study assessed the influence of micro-level streetscape features, including perceptual color, on active school commuting in Hong Kong using SVIs and several ML models (Wu et al., 2023).

Appendix 1 summarizes selected literature investigating the impacts of micro-level street environments primarily using SVI data on various types of active traveling behaviors, including cycling, walking, and running. Coefficient signs of SE characteristics in these studies are also included for later in-depth comparison and discussion with the outcomes drawn from our research. Nevertheless, a detailed explanation of how subjectively measured perceptions (especially less-studied urban design qualities) would locally affect DBS cycling volumes remains poorly addressed.

3. Data & methods

3.1. Analytical framework

The analytical framework comprises five distinct steps to assess the impact of micro-level SE characteristics on DBS cycling, as illustrated in Fig. 1. First, the DBS traffic volume based on a Lime bikeshare trajectory dataset is processed and aggregated at the road segment level (Qiu and Chang, 2020). Second, the micro-level SE characteristics are calculated using SVIs and ML algorithms, including both objective view indices and subjective perceived qualities. Notably, the view index is an objective measure of street features (Ma et al., 2021), while the perceived qualities are subjective senses predicted based on an expert-based visual survey (Qiu et al., 2022). Third, macro-level BE attributes (e.g., accessibility), sociodemographic factors and weather conditions that would influence cycling are collected from census and meteorological data. Fourth, multiple Ordinary Least Squares (OLS) linear regression models are applied to examine the associations between DBS traffic volume and the micro-level SE variables, with other covariates controlled. Additionally, a separate OLS model is used to evaluate weather conditions and their overall impacts on daily DBS volume during the observation period. Finally, a spatial regression model is adopted to capture spatial interactions upon detecting spatial autocorrelation effects, ensuring unbiased interpretations of the relationship.

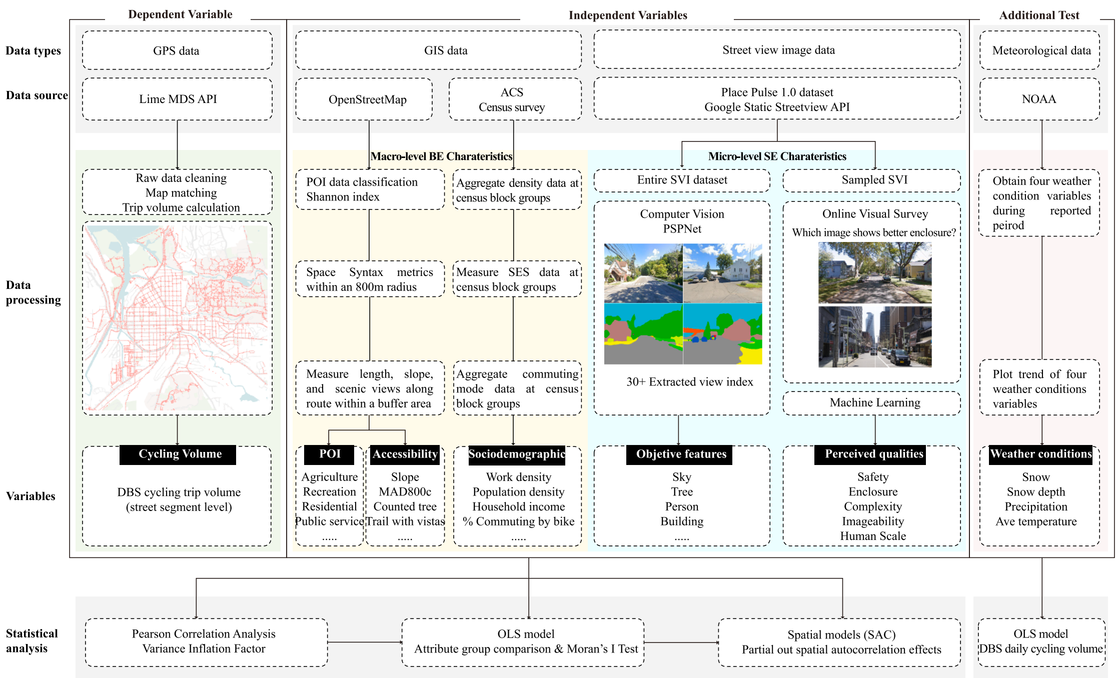

Fig. 1. Analytical Framework.

3.2. Study area

Located in upstate New York, Ithaca is the center of Tompkins County and covers an area of 5.39 miles^{2,} with a population density of ~ 5,571 people/mile^2. It's home to 31,837 residents in 2019, of whom ~ 68 % are white and ~ 50 % are females. Notably, Cornell University has more than 20,000 students residing in Ithaca. The median household income is ~$34,400 (US Census Bureau, 2019). The entire service area of Lime's operation (https://www.li.me/en-co) was set as the study area in this research (Fig. 2).

3.3. Dependent variable – DBS trip volume

Trajectories of Lime DBS trips were collected from the Lime MDS API with support from Lime's local partner, Bike Walk Tompkins (Qiu and Chang, 2020). The raw dataset encompassed 5,038 independent trips from Nov 21, 2019, to Mar 17, 2020 – roughly four months due to data availability. Lime only made route data available after Nov 21, 2019, while its operation was terminated on Mar 17, 2020, due to the impact of COVID-19 (Qiu and Chang, 2020). In Ithaca, Lime had an average ridership of ~ 8,360 monthly trips in 2019, with 80 % being electric (Appendix 2a and 2b). Our trajectory data only represents a small portion of total Lime trips that happened concentratedly in the cold weather (Appendix 2). However, it's still the most comprehensive DBS trip-level trajectory dataset available for our study site.

To ensure data anonymity, no user information (such as name, age, or gender) is identifiable. For each trip, the trajectory data contains a trip ID, a series of coordinates along the route with corresponding timestamps by one-second interval. Appendix 3 provides an example of the raw trajectory records. The cycling trip volume aggregation process is three-fold, encompassing raw data cleaning, map matching, and trip volume aggregation.

3.3.1. Raw data cleaning

First, not every trip record is valid. To ensure data quality, after reflecting on prior related cycling literature on the filtering criteria, we removed rides that were likely to be short "test rides" or "rebalance rides" (Qiu and Chang, 2020): 1) rides had start and end locations less than 0.05 miles apart (i.e., the length of an average Ithaca city block) or 2) rides had a trip duration of less than 3 min (Nasri et al., 2020). The filter left us with 4,430 distinct trip trajectories as the valid dataset for further investigation.

3.3.2. Map matching

Second, due to urban canyon effects, GPS sensors may provide coordinates different from actual routes (Fig. 3(a)) (Chen et al., 2020a). To address this challenge, we need to match every coordinate to the road network shapefile downloaded from Open Street Map (OSM) for each trajectory data (Fig. 3(b)). The road network was split into segments without duplication, yielding 32,649 road segments, each with a unique road ID. We then used a map-matching algorithm that drew perpendiculars from GPS coordinates to road lines. Each coordinate of the trajectory was then matched to the foot of the perpendicular with the shortest length.

$$P_t' = argmin(verticaldis(P_t, R_i)), i\,/in\,\{1, 2, ..., 32649\}$$

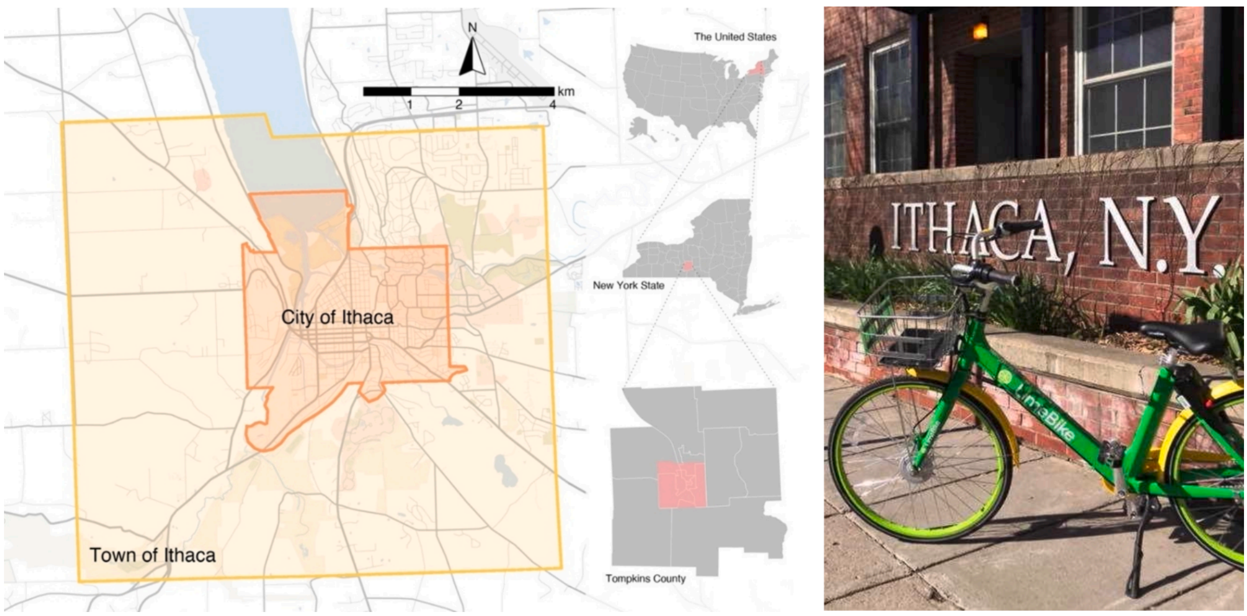

Fig. 2. A) study area and b) its lime dockless bikeshare operation (photo).
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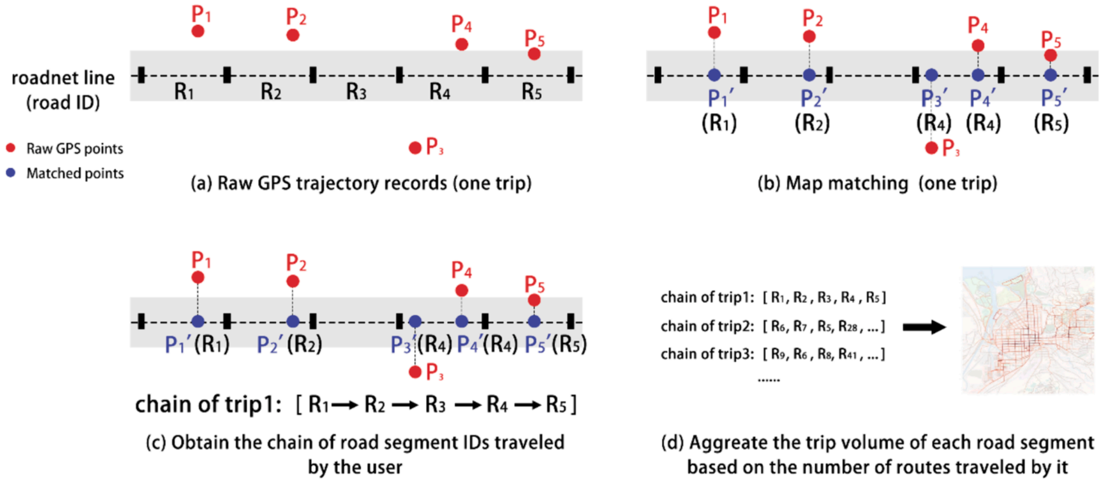

Fig. 3. Illustration of trip volume computation process.

where P_t is the raw GPS point; P_t' is the matched point on the road grid; R_i is the road segment identity; *verticaldis* is the function that computes the vertical distance between the raw GPS point and road segment.

To validate the accuracy of map matching results, we randomly selected twenty routes (Appendix 4) and manually compared raw trajectories with remapped routes. Notably, 90 % (eighteen) of the selected GPS routes were perfectly matched, demonstrating the effectiveness of map-matching framework.

3.3.3. Trips volume computation

Third, we applied Dijkstra's algorithm to reconstruct each route into a series of road segments to aggregate cycling volume (Fig. 3 (c)). Dijkstra's algorithm has been widely used in route chain construction (Dane et al., 2020; Matos et al., 2021; Schweizer et al., 2016). Using Dijkstra's algorithm (see Appendix 5), we identified the shortest path between two matched GPS points (Risald et al., 2017; Salazar Miranda et al., 2021) and bridged the missing pieces of two adjacent matched GPS points with the shortest path. This algorithm returned a set of road segment IDs traveled by the user, facilitated by the *pgRouting* module in PostgreSQL. Finally, we calculated the trip volume of each road segment based on the number of routes traveled by it (Fig. 3(d)).

To this end, we've collected the dependent variable (see Table 1 for a data summary) for later regression analysis. As visualized in Fig. 4, higher DBS cycling activities are spatially concentrated around the Cornell Campus, Collegetown, and Downtown areas. The value of DBS cycling volume varies from 0 to 378, with a mean value of 25.580 and a standard deviation of 40.278 (see Table 2 for descriptive statistics).

3.4. Independent variables

3.4.1. Micro-level SE visual features

Table 1 and 2 provide details and descriptive statistics for all the independent variables. For micro-level SE attributes, SVI data was sampled at a 50 m-interval along each street segment in Ithaca and downloaded from the Google Static Streetview API using each point's corresponding coordinates (Dong et al., 2023; Zhang et al., 2018). The camera settings were consistent throughout all images, with configurations proved to be adequate to represent eye-level perspectives (Qiu et al., 2022; Su et al., 2023; Wu et al., 2023): heading parallel to the street direction, FOV 120 degrees, pitch angle 0 degree, and resolution 600 x 400 pixels. In total, 11,426 valid SVIs were obtained after removing interior or blank views (Song et al., 2023b).

Next, we denoted the percentage of the pixels of a specific visual element in an SVI as the View Index:

$$VI_{fea} = \frac{\sum_{i=1}^{n} PIXEL_{fea}}{\sum_{i=1}^{m} PIXEL_{total}}$$

where VI_{Fea} is the view index of one feature *fea* in one SVI, $\sum_{i=1}^{m} PIXEL_{total}$ is the total pixels of one SVI, and $\sum_{i=1}^{n} PIXEL_{fea}$ is the pixels of streetscape feature *fea* in one SVI.

To effectively extract features and calculate their view indices, we adopted the widely-used Pyramid Scene Parsing Network (PSPNet), a semantic segmentation model whose pixel-level accuracy reaches 93.4 % or beyond (Zhao et al., 2017). Pretrained on the ADE20K dataset, 31 unique categories of objective visual features were quantified for the study area, including ubiquitous features such as the sky and more nuanced features such as trees, bicycles, and streetlights. Such a comprehensive list of objective visual features allows us to enrich existing DBS literature that has only examined limited ubiquitous elements (Wang et al., 2020; Wei et al., 2023; Zhang et al., 2023). Fig. 5(a) to Fig. 5(g) maps the spatial distribution of selected SE visual features, revealing the spatial

Table 1
Summary of all variables in the main models.

Variables	Data Source	Description	Reference
Dependent Variable			
DBS trip volume	Lime Bikeshare	Aggregated to the street segment level	
Independent Variables			
Macro-level BE and Sociodemographic Characteristics			
Sociodemographic Attributes			
Population density (PopDen), work population density (WorkDen)	Census	Aggregated from each census block to the sampling points of the road segments.	
Median Household income (Median_inc), Percentage of people with income below the poverty line (Perc_pover), Percentage of children currently enrolled in primary school (Perc_Prima), Percentage of people commuting to work by carpool (Com_carpoo), Percentage of people commuting to work by public transit (Com_pubtra), Percentage of people commuting to work by bike (Com_bike), Percentage of people walking to work (Com_walk)	American Community Survey	Data is available at the census block group (BG) level in Ithaca, and data from 2020 was used in this study. Aggregated from each census BG to the sampling points of the road segments.	
Accessibility Attributes			
Line connectivity (Lconn)	OSM	The number of other ending lines connected to one line calculated through spatial syntax.	
MAD800c		The mean angular distance within an 800 m radius calculated through spatial syntax.	
Slope (Slope_1)	Thompkins County Data Portal	The slope of the road segment within the buffer area.	
Counted tree (Counted_tr)		The number of street trees in the buffer.	
Cycling infrastructure length (BikeLen_in)		The total length of critical cycling infrastructure within a 50 m buffer.	
Trail with lake (Tr_lake_dmy), Trail with vistas (Tr_vistas_dmy)		Critical recreational and scenic views along the trail as dummy variables.	
POI Attributes			
Agriculture, commercial, industrial, public service (Public Ser), recreation, residential	OSM	The POI data was categorized into seven types, and the number within the 50 m buffer along the road segment was calculated.	
Total Shannon Index (Total_shan)		The mixture of total POI within the buffer area.	
Commercial and residential Shannon Index (C_R_shan)		The mixture of commercial and residential POI within the buffer area.	
Micro-level SE			
Visual features (Objective)			
Ashcan, awning, bicycle, booth, bridge, building, bulletin, car, ceiling, chair, column, earth, fence, fountain, grass, lake, lamp, minibike, mountain, person, pier, plant, railing, road, sculpture, sidewalk, signboard, sky, streetlight, tree, van, wall, water, windowpane	Google SVIs in 600 x 400 pixels	PSPNet was used in Python, and the percentage of the pixels of a specific visual element was calculated.	(Qiu et al., 2022; Song et al., 2022b)
Perceived qualities (Subjective)			
Safety (Q1_Saf)	human perceptions based on the Place Pulse 1.0 dataset	Safety scores were predicted based on the optimal ML model.	(Zhang et al., 2018)
Enclosure (Q1_Enc), Imageability (Q1_Ima), Complexity (Q1_Com), Human scale (Q1_Hum)	human perceptions based on visual surveys reflecting experts' opinions	Four perceived urban design qualities were predicted based on optimal ML algorithms from five tree-based ML models.	(Qiu et al., 2022; Song et al., 2022b)

heterogeneity patterns in the city.

3.4.2. *Micro-level SE perceived qualities*

Perceived qualities were evaluated using an ML-based framework inspired by prior studies (Naik et al., 2014; Qiu et al., 2022; Zhang et al., 2018). In this study, 377 SVIs were randomly selected as visual survey samples to gather scores of four perceived urban design qualities from urban design experts (Zhang et al., 2018), namely enclosure, human scale, complexity, and imageability (Ewing and Handy, 2009; Park et al., 2019). These urban design qualities have been reported to affect various active mobility modes, such as walking (Wang et al., 2022), yet have been rarely (i.e., only enclosure was examined) investigated in the cycling context (Wei et al.,

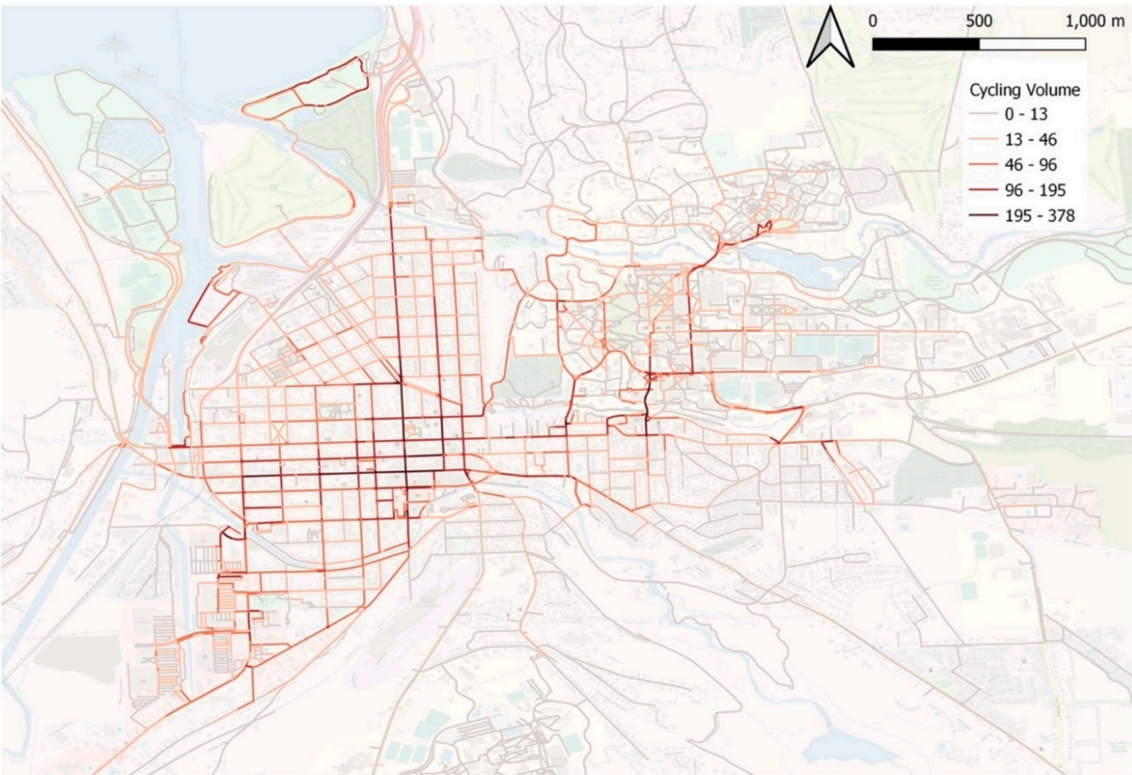

Fig. 4. Spatial distribution of the DBS cycling volume in the city of Ithaca.

2023). Enclosure refers to the sense of confinement, where the streets are defined by vertical elements such as buildings and trees to form a room-like feeling (Ewing and Handy, 2009). Human scale represents streets' physical elements related to the human proportion that contain visual features and details that can be identified, such as the pavement texture, furniture, and plants, that facilitate people's intimacy with the place. Complexity denotes the visually rich environment that holds users' interests, resulting from many noticeable characteristics like building styles, colors, and ornaments, implying the diversity of street usage. Finally, imageability captures the image that the place evokes. It is a culmination of other urban qualities like legibility and transparency, the net effect of distinguishing a place from other areas, making it memorable because of its identity (Park et al., 2019). These urban design qualities are generally more favorable than psychological qualities (e.g., lively) to urban planners and designers as they can inform a more operationalizable method through street transformation (Qiu et al., 2022).

Each expert was asked to select the preferred SVI on an online platform for a pairwise visual survey. In total we recruited 45 participants (male = 25, female = 20) with design backgrounds between the ages of 20–35 and gathered more than four thousand effective clicks. In our case, each image received, on average, 21.5 pairwise comparison votes, exceeding the typical threshold (i.e., 12–36 times) to make the results reasonably reliable (Dubey et al., 2016; Qiu et al., 2023, 2022; Salesses et al., 2013). The Microsoft TrueSkill algorithm was subsequently used to convert the pairwise votes into scores between 0 and 10, with 10 being the most preferable. Using the 377 SVIs, with the four perceived qualities as labels, the view indices were extracted as input explanatory variables to train ML models to predict perceptions in out-of-sample SVIs. It is worth noting that our training sample size is comparable to several related studies when collecting multi-perceptions (Ito and Biljecki, 2021; Qiu et al., 2023; Verma et al., 2020), however, we plan to increase the sample size in the future to further improve the prediction accuracy. Besides design qualities, the widely used Place Pulse 1.0 dataset (Naik et al., 2014) was applied to predict perceived safety, which has been extensively studied in prior cycling literature (Blitz, 2021; Guo and He, 2021; Mertens et al., 2016).

Regarding the ML training procedure, we allocated 80 % of the data for training and used the remaining 20 % for testing. Five tree-based ML algorithms were chosen (e.g., Random Forest) due to their established effectiveness in predicting continuous variables as well as their excellent interpretability (Wang et al., 2019a). The R-squared (R2) and Mean Absolute Error (MAE) were used to compare the prediction results (Table 3). Overall, we achieved an average R2 range of 0.43 to 0.57. Given the training sample size, the accuracy was deemed reasonable compared to prior studies (Cui et al., 2023; Dong et al., 2023; Naik et al., 2014; Qiu et al., 2023). The GB algorithm performed best in predicting three qualities, such as complexity. Notably, imageability exhibited the lowest prediction accuracy, which warrants future research to clarify its definitions and impacts further. Finally, the optimal algorithm (Table 3) was used to predict the subjective perceptions for the rest of the SVIs. The spatial distribution of each predicted perceived quality citywide

Table 2
Descriptive statistics of all variables.

Part 1. Macro-level BE attributes.

Variables	Mean	S.D.	Max	Min	N
DBS cycling volume	25.580	40.278	378.000	1.000	11,426
Sociodemographic Attributes					
Popden	2282.296	4062.659	27896.437	0.000	11,426
WorkDen	2102.251	12831.828	176883.431	0.000	11,426
Median_inc	41944.600	33337.174	99333.000	0.000	11,426
Perc_pover (%)	28.542	16.512	72.989	3.685	11,426
Perc_Prima (%)	3.578	3.442	10.416	0.000	11,426
Com_carpoo (%)	6.679	6.081	22.773	0.000	11,426
Com_pubtra (%)	8.819	7.293	36.152	0.000	11,426
Com_bike (%)	1.511	2.351	14.221	0.000	11,426
Com_walk (%)	27.253	23.597	72.463	0.000	11,426
Accessibility Attributes					
Lconn	2.904	1.122	10.000	1.000	11,426
MAD800c	1045.115	202.209	2898.800	750.332	11,426
Slope_1	3.463	5.265	73.522	0.021	11,426
Counted_tr	7.056	10.937	102.000	0.000	11,426
BikeLen_in	98.818	38.683	291.683	0.515	1533
Tr_lake_dmy	1.335	1.348	1.000	10.000	1212
Tr_vistas_dmy	0.918	0.855	1.000	10.000	1940
POI Attributes					
Agriculture	0.0006	0.025	1.000	0.000	11,426
Commercial	1.478	3.325	29.000	0.000	11,426
Industrial	0.007	0.104	3.000	0.000	11,426
Public Ser	0.021	0.147	2.000	0.000	11,426
Recreation	0.036	0.199	3.000	0.000	11,426
Residential	3.622	6.327	38.000	0.000	11,426
Total_shan	0.244	0.407	2.000	0.000	11,426
C_R_shan	0.151	0.310	1.000	0.000	11,426

Part 2. Micro-level SE, visual features (objective).

Variables	Mean	S.D.	Max	Min	N
Ashcan	0.000005	0.0006	0.025	0.000	11,426
Awing	0.000076	0.001	0.041	0.000	11,426
Bicycle	0.000076	0.001	0.060	0.000	11,426
Booth	0.00000006	0.000006	0.0006	0.000	11,426
Bridge	0.0006	0.008	0.308	0.000	11,426
Building	0.048	0.088	0.971	0.000	11,426
Bulletin	0.0000005	0.00005	0.005	0.000	11,426
Car	0.015	0.030	0.270	0.000	11,426
Ceiling	0.0004	0.012	0.481	0.000	11,426
Chair	0.00001	0.0003	0.022	0.000	11,426
Column	0.00007	0.002	0.179	0.000	11,426
Earth	0.023	0.061	0.884	0.000	11,426
Fence	0.005	0.018	0.373	0.000	11,426
Fountain	0.000004	0.0002	0.018	0.000	11,426
Grass	0.109	0.095	0.706	0.000	11,426
Lake	0.000004	0.0004	0.041	0.000	11,426
Lamp	0.00000003	0.000002	0.0001	0.000	11,426
Minibike	0.00004	0.0008	0.041	0.000	11,426
Mountain	0.002	0.022	0.922	0.000	11,426
Person	0.0004	0.003	0.123	0.000	11,426
Pier	0.00002	0.0009	0.079	0.000	11,426
Plant	0.014	0.036	0.569	0.000	11,426
Railing	0.0007	0.008	0.237	0.000	11,426
Road	0.214	0.116	0.516	0.000	11,426
Sculpture	0.000008	0.0004	0.033	0.000	11,426
Sidewalk	0.018	0.041	0.446	0.000	11,426
Signboard	0.002	0.005	0.130	0.000	11,426
Sky	0.189	0.124	0.488	0.000	11,426
Streetlight	0.0006	0.002	0.035	0.000	11,426
Tree	0.319	0.169	0.997	0.000	11,426

(*continued on next page*)

Table 2 (*continued*)

Part 2. Micro-level SE, visual features (objective).

Variables	Mean	S.D.	Max	Min	N
Van	0.0004	0.004	0.207	0.000	11,426
Wall	0.003	0.015	0.620	0.000	11,426
Water	0.001	0.009	0.304	0.000	11,426
Windowpane	0.00009	0.004	0.351	0.000	11,426

Part 3. Micro-level SE, perceived qualities (subjective).

Variables	Mean	S.D.	Max	Min	N
Q1_Saf	6.826	1.748	9.775	0.702	11,426
Q2_Enc	4.940	0.941	7.502	2.049	11,426
Q3_Ima	4.651	1.042	8.472	1.099	11,426
Q4_Com	3.999	1.063	8.537	1.569	11,426
Q5_Hum	4.933	1.091	8.050	1.087	11,426

is exhibited in Fig. 5(h) to 5(l), with darker colors indicating higher values.

3.4.3. Macro-level BE and sociodemographic attributes

Additionally, the final sets of models controlled various macro-level BE and sociodemographic characteristics to accurately estimate the impact of micro-level SE attributes on DBS trip volume.

POI Attributes: We sourced the POI data from OSM and then classified them into seven categories: Agricultural, Commercial, Community, Industrial, Public Services, Residential, and Recreation. A 50 m buffer was generated to count the total number of POIs for each street segment. Additionally, the Shannon Index was deployed to measure the POI mix regarding all POI functions and the specific commercial-residential ratio. These land use related variables were shown to be significant predictors of cycling behaviors (Faghih-Imani and Eluru, 2015; Shen et al., 2018).

Accessibility Attributes: Two less discussed Space Syntax metrics – Line Connectivity (*LConn*) and Mean Angular Distance in Radius (*MAD*) were deployed to evaluate the accessibility and connectivity of each street segment (Law et al., 2014). The former metric measures how many other lines connect to one line, while the latter determines the mean angular distance within an 800 m radius (i.e., a 5-minute biking distance). Additionally, we considered other critical attributes that impact cycling behavior. For instance, *Slope_l* represents the slope of the road segment, which typically negatively correlates with cycling behavior (Grond, 2016). *BikeLen_in* indicates the total length of cycling infrastructure within a 50 m buffer. *Counted_tr* measures the number of street trees, which informs microclimate control (Dong et al., 2023), while *Tr_lake_dmy* and *Tr_vistas_dmy* are dummy variables that reflect recreational and scenic views, prompting cyclists to choose specific routes.

Sociodemographic Attributes: Population density (*PopDen*) and work population density (*WorkDen*) were obtained from the census block-level data. For each road segment, its value was defined by that of the nearest or belonging census block polygon in QGIS. These two variables had been widely included in previous cycling behavior studies (El-Assi et al., 2017; Faghih-Imani and Eluru, 2016; Noland et al., 2016). Furthermore, many other sociodemographic attributes of cyclists also inform the propensity of cycling, such as cycling prevalence and income level (Grond, 2016; Ogilvie and Goodman, 2012; Wang and Lindsey, 2019). For instance, Wang and Lindsey (2019) revealed that neighborhoods with lower socio-economic status (SES) use bikeshare mode more frequently in the USA. Therefore, utilizing American Community Survey (ACS) data at the census block group level, the study further included median household income (*Median_inc*), percentage of people with income below the poverty line (*Perc_pover*), percentage of children currently enrolled in primary school (*Perc_Prima*) to proxy these characteristics. Additionally, several other variables were included to reflect people's commuting to work using different modes, encompassing the percentage of people commuting by carpool (*Com_carpoo*), by public transit (*Com_pubtra*), by bike (*Com_bike*), and the percentage of those who walk to work (*Com_walk*). Detailed descriptions and statistics can be seen in Table 1 and 2.

3.5. Statistical model specifications and architecture

3.5.1. OLS linear regression

OLS regression has been widely used in urban studies because of its excellent interpretability for policy (El-Assi et al., 2017; Qiu et al., 2023; Su et al., 2023). To select optimal variables for the OLS regression, first, Pearson correlation was used to identify the significant variables that could explain DBS volume, and insignificant ones were removed. Second, we tested the Variance Inflation Factor (VIF) to detect potential collinearity issues among independent variables and variables whose VIF $>$ 10 were removed (Qiu et al., 2023). Third, we fitted several OLS models between each attribute group and DBS volume to examine their explanatory power. Fourth, a baseline OLS model was built with all macro-level BE characteristics (Model 0). Finally, micro-level SE visual features and perceived qualities were added to the baseline model separately as Model 1 and 2 and collectively as Model 3. The performances of OLS models were carefully compared. Table 4 shows the model architecture of these OLS regression models.

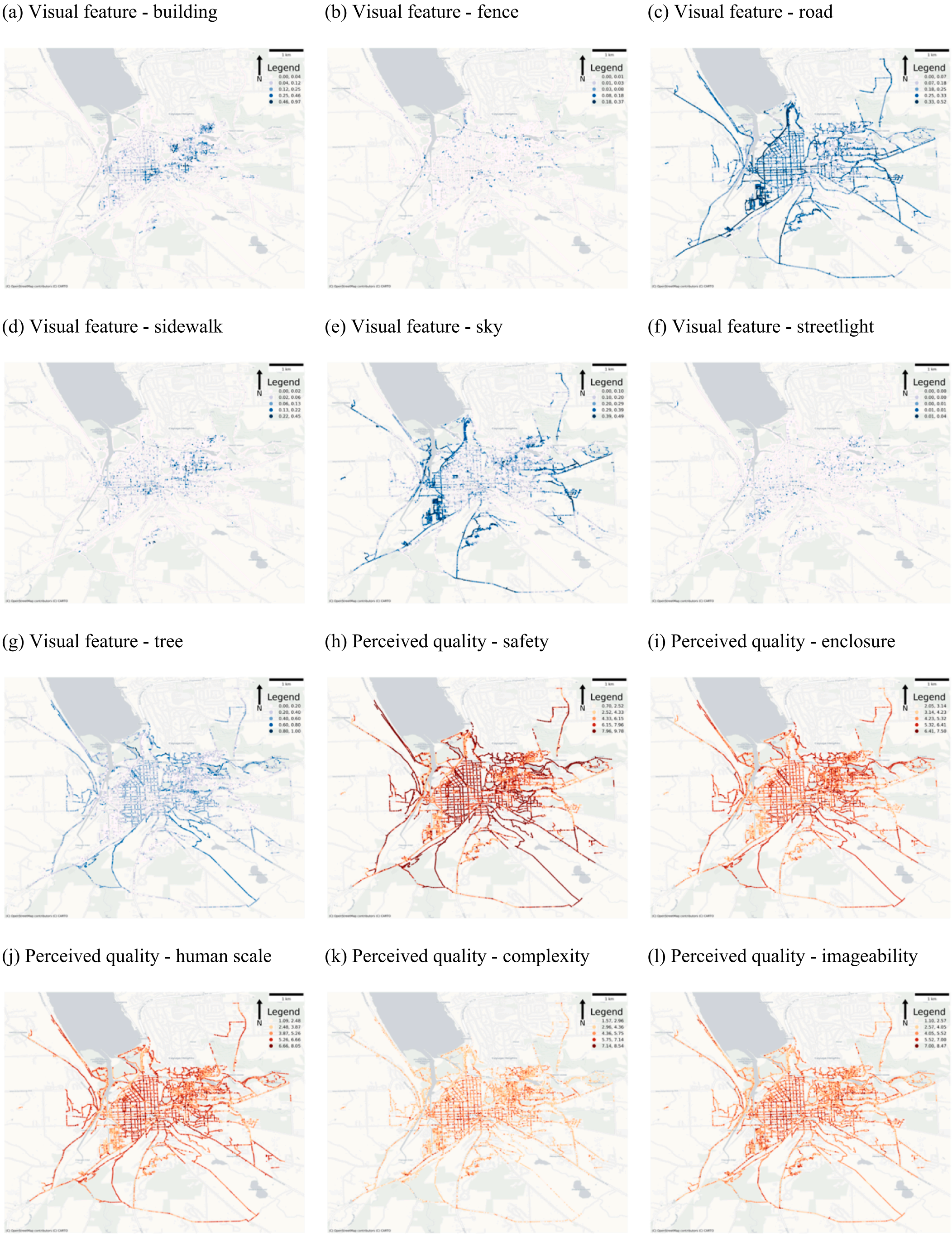

Fig. 5. Spatial distribution of selected micro-level SE characteristics.

3.5.2. Moran' s I test and spatial regression

However, spatial data processes often affect activities in nearby locations, such as cycling and running, leading to spatial autocorrelation and spill-over effects (Dong et al., 2023; Song et al., 2023b). This violates basic assumptions of the OLS and can bias parameter estimates. Therefore, Moran's I statistics were applied to confirm the spatial effects, and Lagrange Multiplier (LM) tests were employed to determine the type of spatial dependency effects to remedy using spatial regression models (Rey and Anselin, 2010).

To mitigate such effects, the spatial lag model (SLM) and spatial error model (SEM) can be employed to properly minimize spatial dependencies. Moreover, when both effects coexist, the Spatial Autoregressive Combined (SAC) Model should be adopted to account for their joint effects explicitly (Elhorst, 2010; Rey and Anselin, 2010). The SAC model (see Eq.(2)) extends the OLS model by including a spatially lagged dependent variable (WY) and a spatially autocorrelated error term (Wμ).

Table 3
Performance of ML algorithms.

	Q1_Saf		Q2_Enc		Q3_Hum		Q4_Com		Q5_Ima	
Model	R2	MAE	R2	MAE	R2	MAE	R2	MAE	R2	MAE
Random Forest (RF)	0.47	1.13	0.57*	1.28*	0.53	1.3	0.47	1.63	0.38	1.79
Decision Tree (DT)	0.27	1.19	0.46	1.46	0.38	1.41	0.26	1.95	0.24	2.01
Voting Selection (VS)	0.48*	1.14*	0.42	1.41	0.53	1.32	0.36	1.62	0.36	1.81
Gradient Boosting (GB)	0.28	1.45	0.45	1.39	0.54*	1.3*	0.49*	1.48*	0.43*	1.62*
ADA Boost (ADAB)	0.32	1.63	0.44	1.48	0.51	1.32	0.36	1.78	0.37	1.82

Notes: * denotes the best performance model.

Table 4
OLS + SAC model architecture.

Model	**Dependent variable (Y)**	**Independent variables (X)**
OLS Linear Regression		
Baseline (Model 0)	DBS trip volume	Macro-level BE characteristics (i.e., POI + Accessibility + Sociodemographic)
Model 1	DBS trip volume	Baseline + micro-level SE "Visual Features"
Model 2	DBS trip volume	Baseline + SE "Perceived qualities"
Model 3	DBS trip volume	Baseline + SE (Visual Features" + "Perceived qualities")
SAC Regression		
Model 4	DBS trip volume	Model 3 + Spatial Lag + Spatial Error term

$$Y = \rho WY + X\beta + \mu \text{ where } \mu = \lambda W\mu + \varepsilon, \tag{2}$$

where ρ (K*1) refers to a vector of the spatial autoregressive coefficient, λ denotes the spatial autocorrelation coefficient. W (N*N) represents the relationship with its neighbors. WY the spatial lag interaction and Wμ captures the interaction effect among the disturbance terms of different spatial units. Y (N*1) a vector of the dependent variable (DBS trip volume), X (N*K) a matrix of the K independent variables (including micro-level SE, macro-level BE and sociodemographic characteristics), β is a K *1 column vector of coefficients of the independent variable, ε (N*1) a vector of independently distributed error terms.

The spatial diagnosis regarding Moran's I and LM tests was conducted using the Pysal package in Python, with the spatial weight matrix W measured using the 'queen' method. The results suggested the appropriateness of using a SAC model to consider both spatial terms. Therefore, based on OLS Model 3, we developed the SAC model as Model 4 (see Table 4) to account for spatial effects.

3.6. Testing the effects of weather conditions on DBS daily trip volume

Much literature has also incorporated weather variables to reveal their impacts (Faghih-Imani et al., 2014; Nosal and Miranda-Moreno, 2014; Shen et al., 2018). For instance, a study in Toronto found that precipitation and snow are barriers to cycling, as inclement weather can lead to increased risks of injuries on the road (El-Assi et al., 2017). Moreover, the temperature often positively correlates with cycling arrival and departure rates, as supported by another docked bikeshare study focused on the Canadian context (Faghih-Imani et al., 2014). Therefore, we also made extra efforts to include several weather variables (i.e., daily average temperature, daily snow, snow depth, and precipitation) to investigate their effects on the overall DBS travel behavior. The detailed methodology used to assess the impacts of weather conditions is explained in Appendix 6, with results reported.

4. Results

4.1. Strength of associations by attribute groups

Based on Pearson correlation results, only variables significantly correlating with DBS cycling volume ($p < 0.1$) were kept for the subsequent analysis. For example, twelve insignificant visual features (e.g., booth, bridge, bulletin) were removed within the SE

Table 5
Model performances by attribute groups.

OLS diagnosis	Sociodemographic attributes	Accessibility attributes	POI attributes	SE Perceived qualities	SE Visual features
No. variables	7	6	9	5	22
R^2	0.213	0.086	0.232	0.108	0.157
Adjusted R^2	0.213	0.085	0.231	0.108	0.156
F-statistic (sig.)	441.8***	178.2***	382.2***	277.4***	96.72***

Notes: p values *** < 0.01.

characteristics. After checking the VIF results, sociodemographic variables like *Median_inc* were further deleted to avoid multi-collinearity. The number of variables of each attribute group kept for final analysis is reported in Table 5.

Table 5 shows the goodness-of-fit (R^2) in explaining DBS volume for the five attribute groups, respectively. They rank as: POI attributes (0.232) > sociodemographic attributes (0.213) > SE visual features (0.157) > SE perceived qualities (0.108) > accessibility attributes (0.086). Notably, the F-statistic tests for all attribute groups were significant ($p < 0.01$). The results imply that the micro-level SE using either measure significantly influences DBS volume, and their explanatory power even outperforms macro-level accessibility attributes, suggesting the necessity of integrating micro-level SE attributes in predicting cycling volumes.

4.2. OLS model performances

The overall performances of OLS models are shown in Table 6. First, compared to baseline ($R^2 = 0.307$), the explanatory power of Model 1 ($R^2 = 0.343$) and Model 2 ($R^2 = 0.328$) is significantly improved by 11.7 % and 6.8 %, indicating the contribution of SE visual features and perceived design qualities. Second, the strength of visual features is slightly higher than perceived qualities, consistent with prior micro-level SE studies (Dong et al., 2023; Song et al., 2022b). Third, Model 3 ($R^2 = 0.354$) demonstrates the highest explanatory power among all OLS models, implying the complementary effect between SE visual features and perceived qualities. In that regard, incorporating subjective and objective measures is more effective than relying solely on either.

4.3. Spatial effects

4.3.1. Spatial clustering of DBS traffic

Nevertheless, Moran's I test over all models' residuals is significant, asserting OLS's inability to consider spatial interaction. The robust LM (lag) and the robust LM (error) statistics also suggest the coexistence of spatial lag and error effects, justifying the use of the SAC model. Besides, the Local Indicator of Spatial Association (LISA) cluster map was also employed to illustrate the spatial distribution of cycling volume in Ithaca (Fig. 6). The hot spots of DBS traffic (High-High clusters) are mainly concentrated in the West End, Downtown, Collegetown, and Cornell Campus. In contrast, the cold spots (Low-Low clusters) are predominantly observed in the urban peripheries.

4.3.2. SAC regression results

Compared to Model 3, the SAC Model exhibits a marked improvement in handling spatial autocorrelation effects (Table 7). Although Moran's I of residuals for the SAC Model remains significant, its magnitude declines significantly from 0.522 to −0.122, with the absolute value close to 0. Furthermore, the explanatory power (Adj. R^2) substantially improves from 0.354 (Model 3) to 0.702 (Model 4), doubling the goodness-of-fit when the spatial lag and error terms are explicitly considered, indicating the effectiveness of capturing spatial interactions to improve the prediction ability and mitigate the bias that existed in the OLS model.

Notably, adding spatial terms also changes the sign, magnitude, and significance of parameter estimates of many variables. This suggests that using the OLS model without accounting for necessary spatial interactions can lead to inaccurate estimations for many BE and SE characteristics. Consequently, our discussion of the impacts of individual variables is based on the spatial model (Model 4).

5. Discussion

5.1. The significance of micro-level SE characteristics

First, both measures, the subjective perceived quality and the objective visual features, matter to the DBS traffic volume. Although the two measures may partially overlap, they collectively exhibit the most substantial predictive power (Model 3), indicating their complementary effects in affecting cycling behavior. This finding extends prior urban studies literature concerning the impact of

Table 6
Overall regression performance of all OLS models.

OLS	Model 0	Model 1	Model 2	Model 3
Attributes	Baseline	Baseline + SE objective features	Baseline + SE perceived qualities	Baseline + SE objective features + SE perceived qualities
R^2	0.307	0.343	0.328	0.354
Adjusted R^2	0.306	0.341	0.326	0.351
F-statistic (sig.)	229.7***	135.1***	205.9***	127.3***
Moran's I on residuals (z-score)	0.553*** (100.646)	0.530*** (96.701)	0.539*** (98.129)	0.522*** (95.273)
Robust LM (lag)	94.691***	266.925***	228.359***	329.268***
Robust LM (error)	203.023***	170.679***	160.210***	165.123***

Notes: p values *** < 0.01, z-scores are shown in the parentheses.

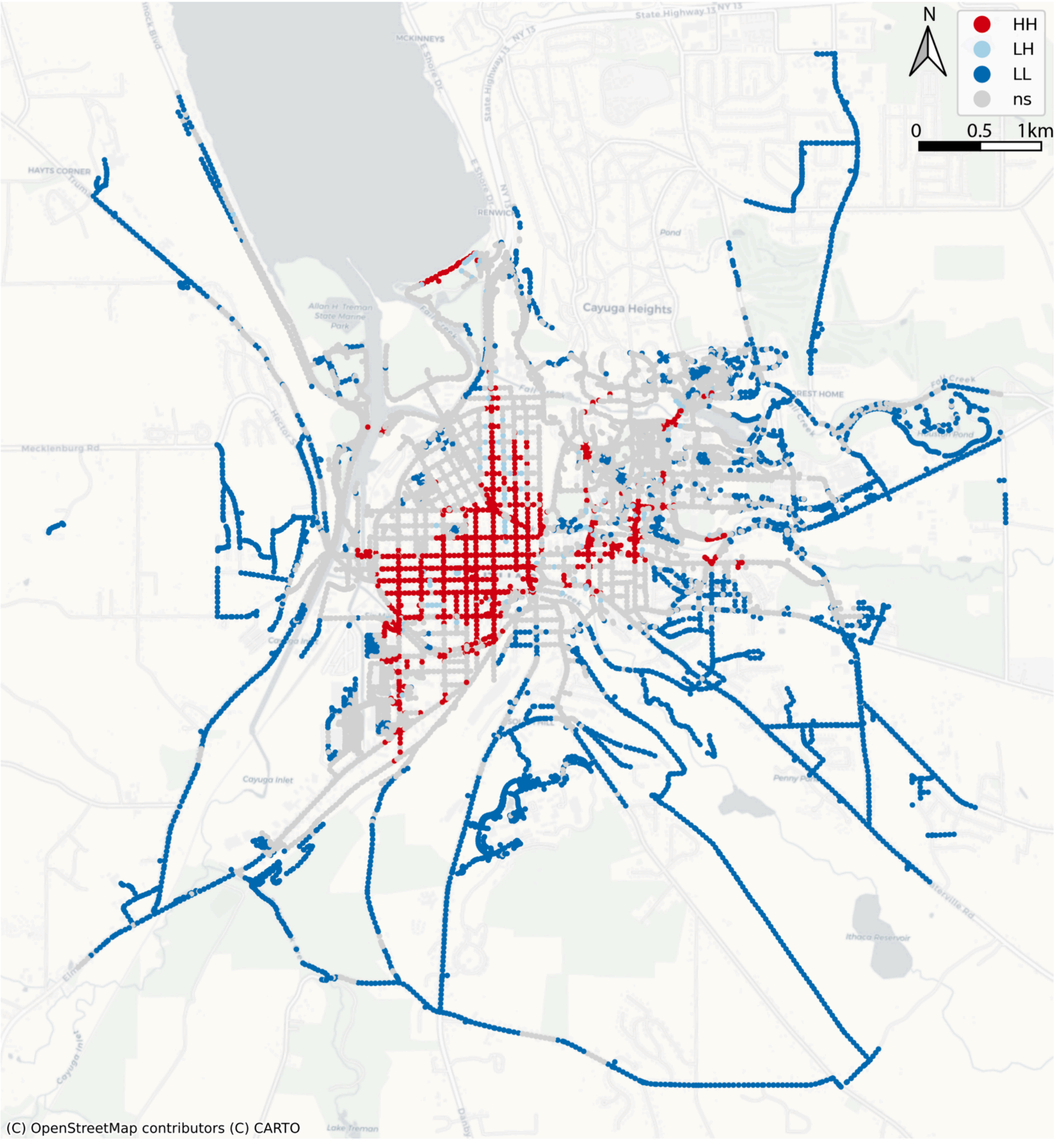

Fig. 6. The LISA cluster map regarding DBS cycling volume.

micro-level SE on housing prices, running, walking to school, and crime rates (Dong et al., 2023; Qiu et al., 2022; Su et al., 2023; Wang et al., 2022; Wu et al., 2023).

Second, the visual features and perceived qualities outperformed the accessibility attribute groups of macro-level BE characteristics but were slightly weaker than the POI and sociodemographic attribute groups (Table 4). This indicates the significant impacts of micro-level SE attributes on DBS traffic and the necessity of including SE characteristics in statistical models for future DBS studies.

Lastly, although model performance improvement was not substantial, implying some overlapping between functions and appearances, SE characteristics could complement macro-level BE in explaining cycling volume (Song et al., 2023b). In addition, the collective explanatory power of both types of SE characteristics (Model 3) is higher than their independent contributions (Model 1 and 2), indicating that while visual features play a more dominant role, adding perceived qualities can offer valuable reciprocal insights. This result supports previous studies demonstrating that overall perceived qualities might capture or represent complicated information that individual visual elements cannot explain (Ewing and Handy, 2009; Qiu et al., 2022). Consequently, the underdressed perceived qualities are as important as visual features in explicating DBS cycling.

Table 7
The results of OLS Models and SAC Model.

Part A. Parameter estimates	Model 0: OLS (baseline- Macro BE + Sociodemographic attributes)			Model 1: OLS (baseline + objective features)		Model 2: OLS (baseline + perceived qualities)		Model 3: OLS (baseline + objective features + perceived qualities)		Model 4: SAC (Model 3 + spatial lag and error terms)	
Variables	VIF	Coef.	Sig.	Coef.	Sig.	Coef.	Sig.	Coef.	Sig.	Coef.	Sig.
Constant	/	0.64		−3.11		−18.551	***	−13.328	**	−12.335	***
Sociodemographic attr.											
PopDen	1.92	0.0001		0.0001		0.0002	***	0.00002		0.0001	***
WorkDen	1.112	0.0002	***	0.0001	***	0.0001	***	0.0001	*	0.00001	
Perc_pover	6.423	−0.045	*	−0.038		−0.054	**	−0.057	**	0.003	
Perc_Prima	4.314	−0.097		−0.16	***	−0.072		−0.159		−0.051	
Com_carpoo	3.948	−0.532	***	−0.468	***	−0.501	***	−0.445	***	−0.068	**
Com_pubtra	3.652	1.255	***	1.096	***	1.213	***	1.081	***	0.087	**
Com_bike	1.985	1.303	***	1.236	***	1.23	***	1.166	***	0.259	***
Accessibility attr.											
Lconn	8.468	2.144	***	1.992	***	1.885	***	1.7179	***	1.188	***
Slope_1	1.994	−0.258	***	−0.053		−0.057		−0.019		−0.038	
Counted_tr	2.612	0.303	***	0.287	***	0.237	***	0.241	***	0.016	
BikeLen_in	2.623	0.045	***	0.06	***	0.058	***	0.065	***	0.027	***
Tra_lake_dmy	2.454	4.909	***	6.216	***	5.918	***	6.857	***	0.742	
Tr_vistas_dmy	2.289	−2.518	**	−0.073		−1.135		0.178		1.058	*
POI attr.											
Agriculture	1.013	−27.36	**	–23.71	*	−29.226	**	−24.865	**	−8.712	
Commercial	2.278	3.537	***	3.024	***	3.195	***	2.846	***	0.382	***
Community	1.875	6.757	***	6.472	***	6.254	***	6.107	***	0.644	
Industrial	1.105	−1.77		−3.662		−3.924		−5.035	*	−1.563	
Public Ser	1.293	4.274	*	4.321	*	4.219	*	3.965	*	−0.43	
Recreation	1.338	7.241	***	8.046	***	7.328	***	7.97	***	3.328	
Residentia	2.517	0.074		0.096		−0.067		−0.033		−0.125	***
Total_shan	7.403	−6.271	***	−7.395	***	−6.646	***	−7.391	***	−2.023	**
C_R_shan	5.446	7.186	**	8.945	***	7.321	***	8.751	***	1.829	
Micro-level SE visual features											
Ashcan	1.022			1208	**			1360.272	***	−77.673	
Awning	1.042			320.84				288.568		−357.074	**
Bicycle	1.019			419.67				323.759		64.775	
Building	2.309			13.585	*			3.628		−1.269	
Car	1.525			2.549				−24.022	**	–33.033	***
Column	1.015			136.14				155.817		147.909	**
Earth	1.624			−24.62	***			−14.908	*	−3.141	
Grass	3.721			−15.73	**			−27.953	***	−8.37	**
Lamp	1.009			116,921				131815.821		72140.582	
Mountain	1.138			−5.671				0.813		3.33	
Person	1.048			468.11	***			215.984	*	157.725	**
Plant	1.276			27.708	***			−38.106	***	−13.477	**
Railing	1.083			69.485	*			82.588	**	36.899	
Road	7.354			51.654	***			38.078	***	21.188	***
Sidewalk	1.59			90.171	***			66.253	***	16.243	***
Signboard	1.149			118.02	*			−16.034		−51.698	
Sky	8.334			−28.01	***			−30.37	***	−3.978	
Streetlight	1.184			610.96	***			499.589	***	−13.444	
Tree	8.545			−4.421				−16.251	**	−1.958	
Van	1.019			179.33	***			134.308	**	84.378	*
Wall	1.101			−11.84				9.393		−10.452	
Water	1.139			36.454				38.069		25.662	
Micro-level SE perceived qualities											
Q1_Saf						0.475		−0.22		0.26	
Q2_Enc						−4.88	***	−2.082	***	−1.513	***
Q3_Hum						3.237	***	3.009	***	1.517	***
Q4_Com						2.785	***	2.265	***	0.59	**
Q5_Ima						3.115	***	2.192	***	0.913	***
Spatial Interaction terms											
W * (trip) (ρ)								0.882	***		
W * μ (λ)										−0.301	

Part B. Goodness of fit	Model 0		Model 1		Model 2		Model 3		Model 4	
R^2(Pseudo R^2)	0.307	***	0.355	***	0.340	***	0.354	***	(0.702)	***
Part C. Spatial dependency										
Moran's I on residuals	0.553	***	0.530	***	0.539	***	0.522	***	−0.122	***

Note: * $p < 0.1$. ** $p < 0.05$. *** $p < 0.01$.

5.2. Effects on DBS traffic

5.2.1. Micro-level SE visual features

The discussion is centered on the SAC model to highlight some interesting patterns. First, the presence of a person exhibits the highest positive correlation among all objective visual features on DBS traffic, contrary to previous studies in dense Asian cities where pedestrians are often seen as impediments to cycling (Guo & He, 2021). This reasonable difference is likely due to lower pedestrian densities in the North American context, reducing the risk of disruptions or collisions (Qiu and Chang, 2021). Our opposite finding is also consistent with the "eyes on streets" theory (Jacobs, 1961), asserting that the constant flow of people could create a sense of security to encourage cycling, especially among certain demographic groups (Ma and Dill, 2017). Surprisingly, despite their recognized role in safety, streetlights show no statistically discernible effect on route volume, possibly due to their low relevance on day trips (Bai et al., 2022; Wei et al., 2023).

Second, as a representation of the classical building styles, the column manifests a positive association with cycling. A possible explanation is that columns enrich the street interface and add diversity, leading to a better cycling experience. As urban design theory asserts, the design of building facades, especially the spatial scale and decoration patterns, are essential to breed activities and shape vibrancy for streets and districts (Montgomery, 1998). Such design features can shape a high-quality public environment and positively influence local perceptions to encourage cycling (Blitz, 2021).

Third, the presence of cars could decrease cycling volume, which concurs with previous research (Blitz, 2021; Grond, 2016; Pucher and Buehler, 2016). It demonstrates that even in small cities, high vehicle traffic continues to be a barrier and can hinder DBS users from cycling due to concerns about being hit by automobiles. Intuitively, lowering perceived car volume and speed could promote cycling behavior, including the odds of cycling and trip duration (Mertens et al., 2016). However, an opposite correlation was found for the presence of vans in this study. One possible explanation is that vans often imply street-level commercial activities and facilities; both as potential destinations are positively related to walkability (Salazar Miranda et al., 2021) and bikeability (Ito and Biljecki, 2021).

Fourth, consistent with prior studies, the road and sidewalk positively affect DBS volume (Zhang et al., 2023). A higher ratio of roads implies wider streets that can better accommodate bike lanes and parking, enhancing bicycle friendliness and raising awareness of perceived bike infrastructure (Ma and Dill, 2015; Mertens et al., 2016). Besides, the sidewalk could suggest a pedestrian-friendly design, which often goes on par with improved cycling infrastructure, thus offering more comfortable and safer routes (Hogendorf et al., 2020; Noland et al., 2023).

Last and unexpectedly, plants (i.e., the low-height flora with leaves and flowers classified by the ADE20K) and grass elements correlate negatively with DBS traffic, while the greenery holds a negative but insignificant impact. However, this finding contrasts with most previous studies where visual greenery positively relates to cycling odds (Lu et al., 2019; Wang et al., 2020) and trip density (Wei et al., 2023), although it aligns with the result of Zhang et al. (2023). One plausible explanation is that dense vegetation may affect the perceived quality of the cycling surface, potentially causing inconvenience to bikers. Another feature included in the model, the awning, also exhibits a negative sign, which can be attributed to its horizontal structure blocking cyclists' sightlines, thereby raising safety concerns and discouraging cycling activity. Nevertheless, these results should be interpreted with caution, underscoring the necessity for future detailed research to test the robustness of our conclusions.

5.2.2. Micro-level SE perceived qualities

First, the impacts of enclosure, human scale, imageability, and complexity are all significant when controlling the spatial effects, except for safety, stressing that urban design qualities effectively influence cycling. The insignificance of safety diverges from numerous previous studies that established a positive correlation between perceived safety and cycling (Bai et al., 2022; Blitz, 2021; Guo and He, 2021; Pucher and Buehler, 2016). Nevertheless, its sign still aligns with our knowledge, suggesting the potential necessity for a larger dataset in future research to comprehend the discrepancy.

Second, most perceived qualities are positively associated with DBS cycling volume, while the impact of the enclosure suggests the opposite direction. Typically, the enclosure is recognized as an attractor for physical activities including walking and running, since it often suggests a sense of safety and vitality (Wang et al., 2019b). However, human perceptions vary depending on different travel modes. The opposite result implies that cyclists might have preferences for street qualities different from walkers and runners. Interestingly, this negative relationship aligns with a previous walking behavior study in an auto-oriented city in the USA (Park et al., 2019), thereby warranting further exploration.

Third, the human scale shows a positive correlation, outstripping imageability and complexity by 60 % and 153 % in magnitude, underscoring the importance of human-oriented street designs. Both imageability and complexity exhibit positive correlations with DBS cycling volume. Every one-point increase in imageability corresponds with roughly one additional DBS trip, and its impact surpasses complexity by about 57 %. These two qualities, by definition, share some common components, such as architectural diversity, landscape features, and human activity, thus implying that a place with visual richness can enhance aesthetics, attract attention, and leave a memorable impression, ultimately promoting DBS cycling. It largely echoes prior studies that emphasized the role of perceived aesthetics in boosting cycling volume and physical activities (Blitz, 2021; Pucher and Buehler, 2016; Zhang et al., 2023).

5.2.3. Macro-level BE variables

In addition to micro-level SE characteristics, the macro-level BE still matters. On the one hand, among POI attributes, commercial and community land use positively influence cycling volume, while residential land use shows a negative correlation, consistent with previous studies (Wang and Noland, 2021). This supports prior research on bike-sharing usage in areas near restaurants, parks, and stores/shops (Faghih-Imani and Eluru, 2015), underscoring the importance of non-residential land use for DBS users. However, it contrasts with studies in China, where retail shops had negative correlations with cyclists, possibly due to geographical differences (Wang et al., 2020). Surprisingly, the Shannon mix entropy score shows no significant impact on cycling volume, contradicting many studies linking mixed land use with cyclists (Faghih-Imani & Eluru, 2015; Lu et al., 2019). Our assumption might be partially supported by H. Wang & Noland (2021)'s evidence indicating a minimal association during weekdays, yet a positive correlation on weekends, calling for further investigation on temporality.

On the other hand, within accessibility attributes, trail with vista, street connectivity, and length of bike lanes disclose positive impacts. Previous research has revealed that cyclists are drawn to aesthetically appealing surroundings, such as beautiful greenery and pleasant streetscapes (Blitz, 2021). The positive effects of trails with vistas indicate that cyclists prefer routes that provide picturesque views. Additionally, prior studies highlighted that street connectivity is vital for cycling because it enhances efficiency. Bike lane provision is essential for promoting cycling, given their safety benefits, particularly when situated alongside busy roads (Pucher and Buehler, 2016). Interestingly, the number of street trees shows an insignificant relationship with the cycling volume, as supported by Lu et al. (2019)' s finding on the impact of overhead greenness. Additionally, the insignificant negative coefficient of the slope suggests that although the topography can deter cyclists, its effect might be minor for DBS users in Ithaca.

5.2.4. Sociodemographic attributes

First, our analysis indicates that population density shows a positive effect, while the work population density is insignificant. This aligns with previous observations suggesting that dense urban design in Western countries generally promotes cycling (Ding and Gebel, 2012; El-Assi et al., 2017) while discouraging cycling in high-density Asian cities (Lu et al., 2019), possibly due to differences in cultural and historical contexts affecting urban mobility dynamics.

Second, our model offers partial support to Wang and Lindsey (2019), who noted that neighborhoods with lower SES profiles are associated with higher cycling trip volume. In this research, the percentage of people below the poverty line reveals a positive impact, although with an insignificant p-value. Additionally, it is no surprise that the percentage of people commuting by bike shows a significantly positive effect on DBS volume, with its strength approximately three times that of people commuting by public transit. This contributes to the ongoing debate on whether the public bikeshare system complements or competes with public transit (Griffin and Sener, 2016; Martin and Shaheen, 2014; Qiu and Chang, 2021). However, it is worth noting that to increase the adoption of bikeshare for commuting, workplaces also require facilities like bike parking and showers (Buehler, 2012).

Lastly, carpooling to work negatively affects Ithaca's cycling volume. This suggests that while carpooling and bike-sharing are shared mobility options contributing to green transportation, they might cater to different SES groups and serve different geographic areas, particularly regarding travel distances to workplaces.

5.2.5. Weather variables

Our separate statistical model (see Appendix 6), fitted to daily trip volume, reveals insightful findings regarding weather conditions. Consistent with existing literature(Faghih-Imani et al., 2014; Nosal and Miranda-Moreno, 2014), we found that average temperature positively influences trip volume ($\beta = 0.784$), indicating that higher temperatures tend to increase cycling, and its effect is possibly more pronounced in winter than in summer. This finding is particularly noteworthy in the context of a smaller North American city, where such data is often underrepresented in research.

As expected, precipitation ($\beta = -21.003$), snow ($\beta = -2.386$), and snow depth ($\beta = -1.007$) all exhibit a negative sign with varied magnitude. However, their parameters fail to reject the null hypothesis. Thus, this dataset cannot confidently draw a solid conclusion. That said, the insignificance of precipitation aligns with Ashqar et al. (2019) but contrasts with other literature (El-Assi et al., 2017;

Nosal and Miranda-Moreno, 2014; Shen et al., 2018), potentially due to our data being predominantly from winter months and lacking variability. The ADF test result elucidates their stationarity, thus supporting this speculation. Lastly, it is worth noting that examining the effects of weather conditions on the spatial distribution of DBS trips is beyond the scope of this article, and we discuss this in the limitations section.

5.3. Comparing impacts of micro-level SE across active transportation modes

Our findings further enrich the understanding of micro-level SE's varied effects on the three micro-mobility modes: walking, running, biking, and across various cycling services (i.e., docked bikeshare, dockless bikeshare, personal bike). First, regarding objective visual features, only a few streetscapes, i.e., roads, sidewalks, and people, positively influence all three modes consistently, while the effects of ubiquitous natural elements diverge considerably. Remarkably, the sky and water contribute to walking and running in metropolitan areas (Dong et al., 2023; Sevtsuk et al., 2021) while they are less influential on DBS cycling in Ithaca. Plants and grass, typically neutral for running and walking, can hinder cycling in Ithaca, likely because they compete for the share of street space with sidewalks and bike lanes in a small city setting with limited public funding and narrower streets.

Interestingly, cars, often conceived as an impediment to all cycling service types because of crash risks (Blitz, 2021; Pucher and Buehler, 2016), can be insignificantly negative or even sometimes positively correlated with walking (Basu and Sevtsuk, 2022; Wu et al., 2023) and running behaviors (Dong et al., 2023; Jiang et al., 2022a). It suggests that, compared to cyclists, pedestrians are more adaptive to car-oriented neighborhoods and inexorably share streets with cars in urban areas. Moreover, subtle elements like trees and street furniture (e.g., ashcans, awnings, streetlights, walls, and signboards) influence running and walking (Dong et al., 2023; Salazar Miranda et al., 2021) but are less significant for DBS cycling in Ithaca, likely due to the higher travel speed of cyclists and fewer interactions with such features. Nevertheless, in high-density metropolitan cities, street trees and visual greenery tend to positively influence cycling consistently, regardless of service types (Lu et al., 2019; Wang et al., 2020; Wei and Zhu, 2023; Wei et al., 2023).

The effects of perceived qualities also vastly diverge by mode. In Ithaca, enclosure negatively impacts DBS cycling, whereas complexity poses a positive effect. This starkly contrasts conclusions drawn on other modes in larger metropolitan cities. For example, enclosure tends to promote walking (Wang et al., 2022), and complexity deters walking route deviations (Salazar Miranda et al., 2021). Such discrepancies may be partially explicated by cyclists' active engagement with the street environment (Ma et al., 2014). Furthermore, cyclists, irrespective of the cycling service type, generally favor high-quality and comfortable public spaces (Blitz, 2021; Mertens et al., 2016; Zhao et al., 2022), converging with positive directions in our study on design qualities such as imageability and human scale, suggesting certain homogeneity in perceptions within the cycling mode. This implies that interventions for one cyclist group could resonate well with others, highlighting the need for tailored urban design strategies that can benefit various groups collectively. Finally, perceived safety emerges as a universally important determinant across all modes and cycling service types, highlighting its importance in running (Dong et al., 2023), walking, DBS cycling (Guo and He, 2021), docked bikeshare cycling (Bai et al., 2022; Zhao et al., 2022) and personal biking (Blitz, 2021; Ma et al., 2014; Mertens et al., 2016). Such strong consistency emphasizes its critical role in urban planning to satisfy the basic needs of people in the city (Lynch, 1960; Ramírez et al., 2021). These comparisons further underscore the similarities and differences in impacts based on travel modes and urban contexts. Sidewalks, for instance, enhance cycling and walking in Western cities but may discourage them in developing countries due to obstructions like street vendors (Wang et al., 2022; Zhang et al., 2023).

The convergences and divergences across modes, city sizes, and socioeconomic statuses justify further investigations of the micro-level SE characteristics. Scholars can also apply our analytical framework to other less-understood cycling service types like docked bikeshare or other modes for more in-depth inferences. Our findings suggest a paradigm shift towards a collaborative approach, where transport planners and urban designers work together to create street environments that support diverse modes of travel while considering the potential synergistic and antagonistic effects of various interventions (Goodman et al., 2014).

5.4. Implications for urban planning and design

Our findings equip urban planners and designers with practical knowledge to prepare policy and design interventions that foster a bike-friendly street environment. DBS users in small cities like Ithaca favor rides on streets designed to the human scale, characterized by enhanced visual appeal, memorable city image, wider roads and sidewalks, and correspondingly serving more pedestrian flows. Conversely, streets with higher enclosure or more cars may adversely affect the cycling experience, becoming less preferred by DBS cyclists. These findings suggest that planners should broaden their focus from macro-level BE interventions to the micro-level SE, where changes are more noticeable and cost-effective (Stanislav and Chin, 2019). Previous studies suggested that defining a roadmap to enhance perceptions of micro-level SE directly can help foster a virtuous cycle to encourage cycling trips (Ma et al., 2014), with attention paid to transforming the holistic perspective of SE instead of individual elements (Zhao et al., 2022).

Specifically, alongside the long-term improvements in road network accessibility and land use mix, urban designers have a role in creating more expansive, open streets with various building styles to enhance visual diversity. Some interventions can be temporary and cost-effective. For example, the temporary introduction of outdoor curbside dining patios during the pandemic shows how the SE can become more human-centric to encourage cycling (Mandhan and Gregg, 2023; Song et al., 2024). Movable planters with perennial plantings can physically demarcate patio spaces from cycling lanes, replacing underutilized lay-by parking spaces that might lead to conflicts between cyclists and vehicles. Such tactical transformation experiments within the SE can stimulate neighborhood cycling behavior (Noland et al., 2023). These measures aim to reclaim the streets from traffic, creating a unique identity and uplifting the place's imageability, eventually invigorating the street's vitality (Jiang et al., 2022b; Park et al., 2019). Conversely, enclosed streets with frequent interruptions or obstructions, such as improperly placed plants and overgrown grass, can discourage cyclists. Consequently, government officials can strategically incorporate new cycling facilities in neighborhoods with good visibility and visual openness. Simultaneously, they can implement traffic calming measures to reduce vehicular speeds, ensuring safety for DBS users and encouraging DBS travel demand (Pucher & Buehler, 2016). Such steps promote more sustainable, resilient, and livable neighborhoods to achieve the planning goal of 'streets for people' in the long term (Bertolini, 2020).

6. Conclusion and limitation

6.1. Conclusion

This study quantified DBS trip volume at the street segment level in Ithaca using trajectory data from Lime bike users to delve into the relationship between trip volume and the overlooked micro-level SE characteristics. Notably, this research extends the discussion to compare the influences of micro-level SE on DBS cycling with other active travel modes (e.g., walking and other forms of cycling) to inform urban planning interventions that can synergistically enhance active travel across different modes.

These findings underscore the critical influences of both objective visual features and perceived qualities on DBS users. First, introducing micro-level SE characteristics significantly improves the explanatory power of the baseline model, which consists of macro-level BE and sociodemographic variables. Notably, accounting for spatial autocorrelation effects demonstrates a non-negligible improvement in model fit.

Second, the impacts of micro-level SE surpass those of many macro-level BE variables. The objective visual features (e.g., green view), whether assessed individually or jointly, tend to exhibit a more substantial influence on DBS volumes than perceived qualities. Moreover, incorporating both objective visual features and subjective perceived qualities outperformed all other models, suggesting that perceived qualities complement visual features to holistically describe the micro-level SE. Specifically, we found that streets that are wider, open, and visually perceived as more complex, human-scale, and imageable, attract more DBS trips. Conversely, streets characterized by a higher degree of perceived enclosure or excessive obstructions (e.g., cars and grass) may become barriers for DBS riders.

Finally, compared to other modes (i.e., walking and running) and cycling types (i.e., docked bikeshare and personal bike), we found several objective features (e.g., roads, sidewalks) positively influence all three active modes. Regarding perceived qualities, different cycling types consistently favor places with good imageability and human scale, and safety is viewed as a universally important factor across all scenarios. On the other hand, natural (i.e., sky and water) and subtle (i.e., trees and street furniture) objective features play a more influential role in walking and running than cycling because of cyclists' higher travel speeds. Moreover, cars, plants, and grass may deter cycling, while they are typically not a significant predictor for walking and running. Similarly, perceived enclosure negatively impacts DBS traffic but tends to encourage walking. On the contrary, perceived complexity discourages walking while promoting cycling.

In summary, relating our findings to other active travels provides invaluable insights and cost-efficient street design measures for urban planners, urban designers and transport planners on micro-level SE interventions that will foster more active travels. For example, urban cycling routes should be designed to be more complete, human-scale, open, visually complex, and intriguing, further enriched by diverse building types to create a memorable place with good imageability. Moreover, the framework can be applied to various contexts, e.g., to identify locations that could stimulate cycling activities and to pinpoint locations where the actual cycling traffic disagrees with the conventional bikeability evaluation (Ma and Dill, 2017).

6.2. Limitations and future work

Although this study is highly replicable to other cities to inform how micro-level SE attributes affect active travel, we acknowledge five areas for future improvements concerning (1) the omission of temporality, including weather conditions; (2) the ignored gender difference in perceived visual qualities; (3) the incapability to infer causality; (4) the lack of comparison with other active modes like scooter sharing; and (5) the potential improvement of statistical methods.

First, weather conditions are significant predictors of cycling trips in space and time (Corcoran et al., 2014), warranting more sophisticated investigations if full-year trajectory data is available. Although this study has investigated the association between weather and daily DBS traffic, no evidence can be drawn, possibly due to the lack of variability in weather throughout the observed period. Meanwhile, although weather conditions might determine daily trip volume with a lagged effect (Nosal and Miranda-Moreno, 2014), the route decisions of DBS users are still more likely to be informed by the micro-scale SE attributes once the trip generation and mode choice decisions are determined. Additionally, we acknowledge that DBS trip patterns can be temporally dynamic (Shen et al., 2018). For example, Qiu & Chang (2021) found more bus-bikeshare linkage trips happened on weekdays than weekends in Ithaca. Future studies in this regard should investigate how the roles of SE characteristics on DBS volume could diverge by day in the week and time in the day.

Second, there is a disagreement regarding whether socio-demographic factors like gender may influence the perceived visual qualities of the same street environment. Future studies can investigate if there are non-negligible divergences being attributed to gender regarding the perceived visual quality of street scenes (Ramírez et al., 2021). On the one hand, based on 1.5 million responses from the internet, Salesess et al. (2013) found no significant variation in people's perceived safety across socio-demographic groups. On the other hand, Ma and Dill (2017) asserted that women with children tend to perceive a highly bikeable street as low bikeability using self-reported surveys. Similarly, women are more likely to underestimate the safety of a place due to the fear of crime (Cui et al., 2023). Given the gender-related disagreements in perceived safety, future studies can differentiate the gender of DBS users to better understand the role of socio-demographics on route choices for active travel.

Third, although this study manifested the correlations between micro-level SE attributes and DBS traffic, no causal relationship can be inferred. The results should be interpreted with caution given the fact that there are possible reverse causations (e.g., the street environment is improved as a result of more cycling traffic by urban designers). Future research in this regard could explore the underlying causal mechanism with panel data when available.

Fourth, although we compared the impacts of micro-level SE attributes across different active travel modes, scooter sharing is not discussed and compared due to the scarcity of relevance in the literature (Jin et al., 2023). Nevertheless, emerging studies indicated that scooter-sharing trips generally travel longer distances, are primarily purposed for recreational trips, and are less influenced by weather than bike-sharing trips (Liu and Lin, 2022). Therefore, it is reasonable to assume that the ease of mobility and comfort from scooter sharing could facilitate more subtle interactions between scooter riders and the micro-level SE, while perceived traffic safety can become an imminent concern pertaining to scooter riders due to its faster travel speed (Kang et al., 2024).

Last, more advanced statistical methods can be deployed to unravel the potential non-linear roles of the micro-level SE attributes, avoiding the oversimplification (e.g., linear assumption) of the complex travel behavior of active modes (Wei et al., 2023). Moreover, alternative methods such as Factor Analysis (FA) can offer a fresh perspective to strategically reduce the feature space dimensions while keeping the most valuable meanings by recombining variables, allowing for interpretations of their interaction effects (Song et al., 2022a).

CRediT authorship contribution statement

Qiwei Song: Writing – review & editing, Writing – original draft, Visualization, Software, Investigation, Formal analysis, Data curation, Conceptualization. **Yulu Huang:** Writing – original draft, Visualization, Software, Investigation. **Wenjing Li:** Writing – review & editing, Data curation, Conceptualization. **Faan Chen:** Writing – review & editing, Validation, Supervision. **Waishan Qiu:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix

A1. Selected literature regarding the impacts of SE on active travel

General info			Y (Dep. Var.)		X (Ind. Var.)			Models and findings	
Paper	**Mode**	**Region**	**Measures**	**Data Source (scale)**	**Streetscape features**	**Perceptions**	**Other variables**	**Method**	**Findings and limitations**
X. Wang et al. (2022)	Walk	Central Beijing	Odds of walking to school	3,357 trip records from travel survey (TAZ)	N/A	**Positive:** enclosure **Negative:** walkability, street safe facilities	Macro-level BE characteristics	Generalized additive mixed models (GAMMs)	**Limitation:** 1) Ignore individual street feature; 2) Data doesn't reflect street-level walking behaviors
Salazar Miranda et al. (2021)	Walk	Boston	Desirability index	28,363 GPS trajectories (Street segment)	**Positive:** urban furniture index, sidewalk index	**Positive:** visual enclosure index; **Negative:** facade complexity index	Macro-level BE characteristics	OLS model	**Limitation:** Only investigated route deviation from shortest routes.
Sevtsuk et al. (2021)	Walk	Boston	Pedestrian route volume	14,760 GPS trajectories (Street segment)	**Positive:** sky view factor **Insignificant:** green view index	N/A	Length, Turns, sidewalk width, speed limit, traffic volume	Path size logit (PSL) model	A 10 % increase in sky view leads to 8.1 m of walking distance. **Limitation:**1) limited objective features; 2) ignored perceptions
Dong et al. (2023)	Run	Boston	Running trip volume	26,201 trips from Strava Heatmap (Street segment)	**Positive:** sky, road, sidewalk, person, wall, water; **Negative:** traffic light, ashcan	**Positive:** safe, wealthy **Negative:** lively	5Ds (e.g., density)	OLS model, SAC model, GWR model	R2: BE (0.180) > SD (0.138) > SE (0.127) **Limitation:** Perceived qualities only included psychological perceptions.
Jiang et al. (2022)	Run	London	Running trip volume	48,286 trips from Strava Heatmap (Street segment)	**Positive:** tree, road, sidewalk, signboard, streetlight, water, booth, person, car, awning, mountain; **Negative:** wall, fence, plant, bridge	N/A	5Ds + Sociodemographic factors	OLS model, SAC model	The adjusted R2 values of the macro-scale 5Ds within three buffers and micro-scale were 0.365, 0.372, 0.373, and 0.146. **Limitation:** ignored perceptions
Ettema (2016)	Run	Netherlands	running frequency	4,395 surveys (buffer zone)	**Positive:** cyclists **Negative:** threats	N/A		OLS model	After adding subjective variables, R2 improved. **Limitation:** 1) no real-time traffic volume; 2) limited sample size; 3) ignored visual features
Ma et al. (2014)	Personal bike	Portland	Cycling frequency	830 observations from phone surveys	N/A	**Positive:** accessible, easy, safe	Macro-level BE + demographic attributes	SEM	Perceptions: direct impact; BE: indirect. **Limitation:** no visual features; limited perceptions
Lu et al. (2019)	Personal bike	Hongkong	Odds of cycling	5,701 trips from surveys	**Positive:** street greenness	N/A	Macro-level BE + demographic attributes	OLS model	**Limitation:** 1) only one visual feature; 2) ignored perceptions
Blitz (2021)	Personal bike	Offenbach am Main, Germany	Cycling intention and attitude	701 cases from household surveys	**Positive:** perception of safe cycling infrastructure; greenery; **Negative:** vandalism, dirt, and high car pressure	**Positive:** high quality public space	Sociodemographic + travel mode availability	Binary logistic regression	**Limitation:** 1) limited visual features; 2) limited perceptions
Zhao et al. (2022)	Docked bikeshare	Beijing	feeder mode & travel route choice	403 commuters from Travel Survey	**Positive:** tree shade (positively correlated to people with higher income or higher education)	N/A	Socioeconomic factors + travel distance	Binary logit model	**Limitation:** 1) limited visual features; 2) ignored perceptions; 3) no real bike trajectories

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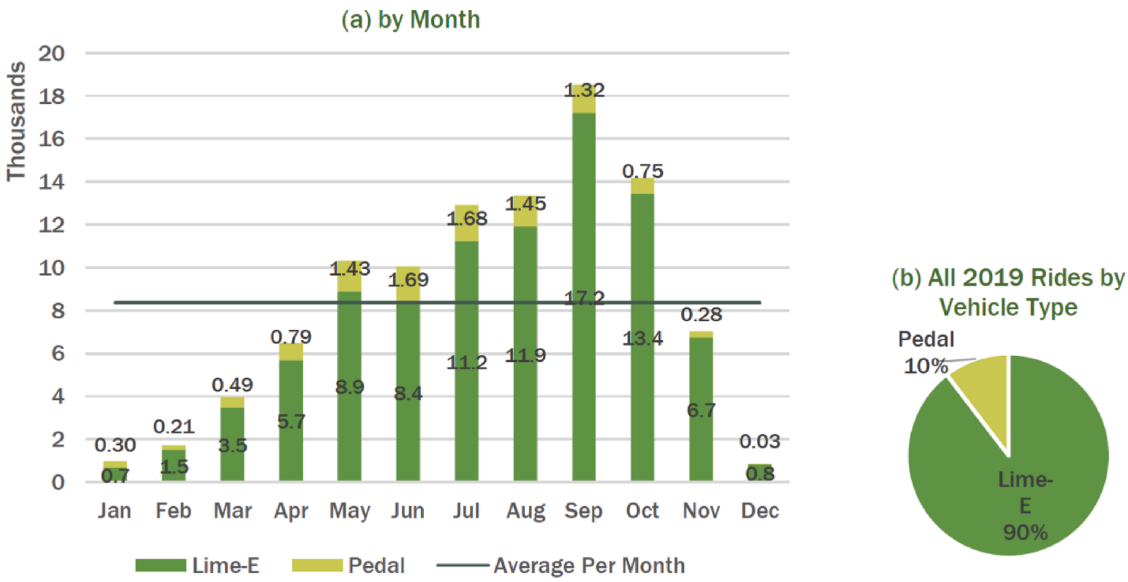
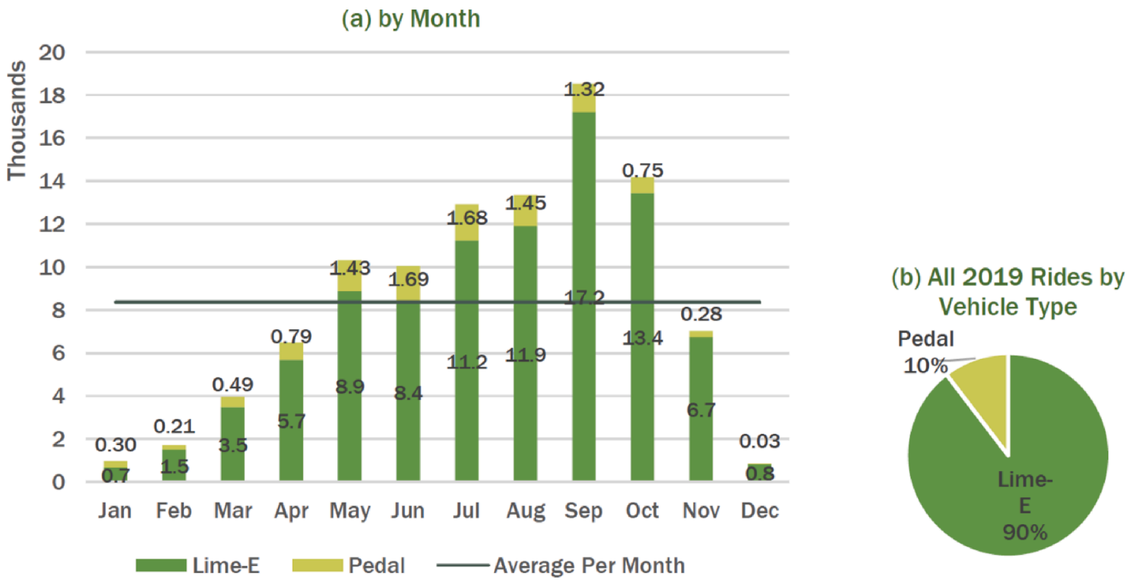
(*continued*)

General info			Y (Dep. Var.)		X (Ind. Var.)			Models and findings	
Paper	**Mode**	**Region**	**Measures**	**Data Source (scale)**	**Streetscape features**	**Perceptions**	**Other variables**	**Method**	**Findings and limitations**
Bai et al. (2022)	Docked bikeshare	Xi'an	Bicycle usage levels	2918 samples from surveys (1 km buffer)	**Positive:** presence of streetlights (residential); **Negative:** lack of bicycle parking sheds (workplaces)	**Negative:** safety concerns* (residential)	Sociodemographic factors + travel habits + travel attitudes	Multinomial logistic models	**Limitation:** limited visual features and perceptions
B. Wei & Zhu (2023)	Docked bikeshare	New York	Competition, integration, etc.	998 docking stations data (200 m buffer zone)	**Positive:** street trees** (positively related to competition mode)	N/A	Social economy + transportation systems + land use	OLS model, SLM model	Users in competitive mode value the sightseeing and leisure benefits. **Limitation:** 1) limited visual features; 2) no perceptions 3) limited sample sizes
E. Zhang et al. (2023)	DBS	Beijing	distance, speed & volume	16,266 trips (street segment)	**Positive:** road; **Negative:** pavement	N/A	5Ds + physical disorders	OLS model	**Limitation:** limited visual features but no perceptions
R. Wang et al. (2020)	DBS	Shenzhen	Cycling frequency	Trips aggregated to 167 metro stations	**Positive:** street view greenspace***	N/A	Macro-level BE	Multivariate Poisson regression	**Limitation:** only considered one feature but no perceptions
Wei et al. (2023)	DBS	Beijing	DBS-metro transfer trip density & speed	567,364 access transfer and 581,798 egress trips (500 m buffer)	**Positive:** greenery, presence of barriers; **Negative:** presence of streetlights, presence of traffic signals	N/A	Macro-level BE + socioeconomic + metro station attribute	Multiple linear regression model	VIF > 10: openness, building enclosure. **Limitation:** very few visual features; no perceptions

Note: variables that show insignificant effects were not included in this table.

A2. Descriptive summary of the Lime data




(a) by Month
Thousands
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18
16
14
12
10
8
6
4
2
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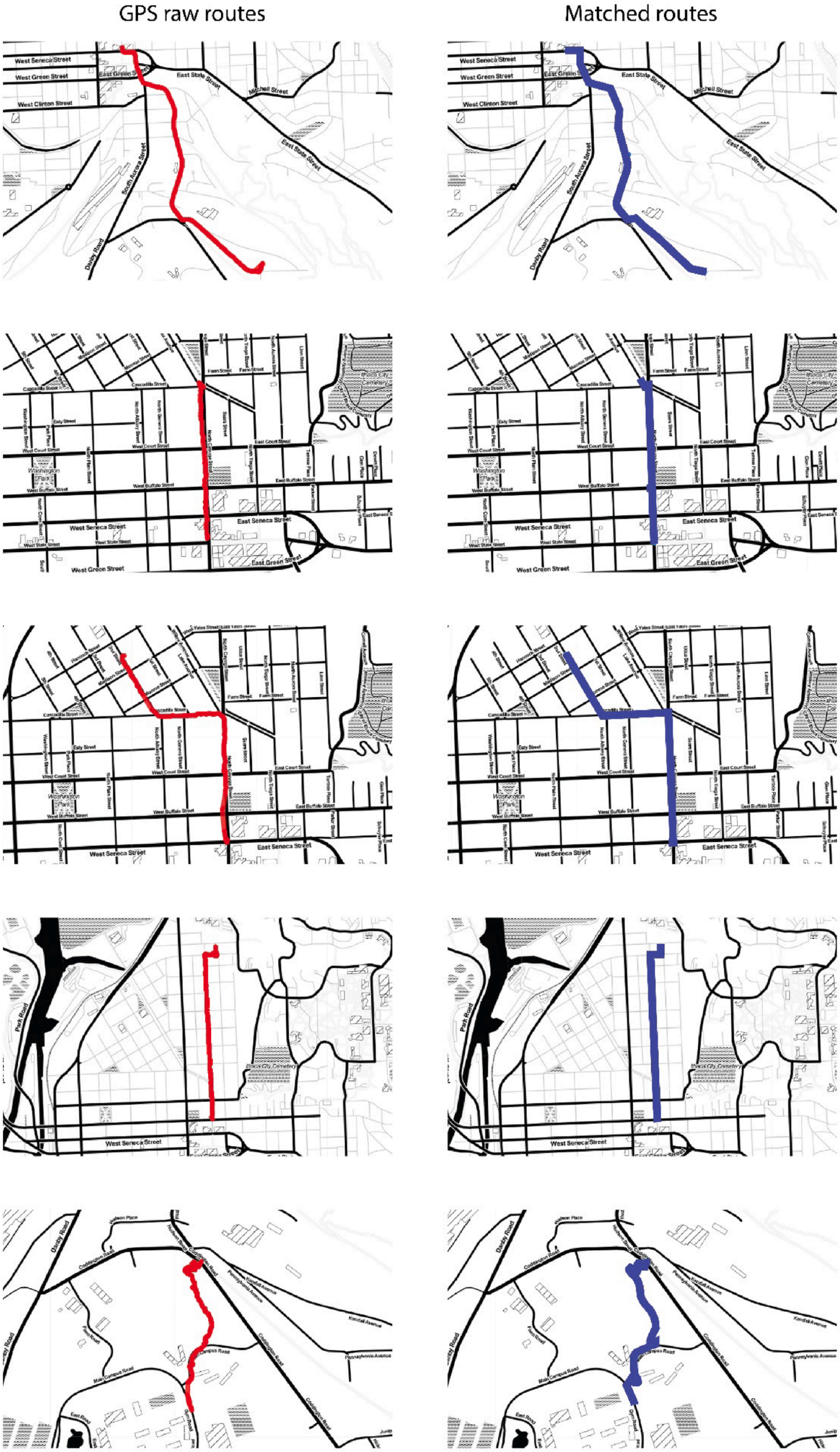
0.30
0.7
0.21
1.5
0.49
3.5
0.79
5.7
1.43
8.9
1.69
8.4
1.68
11.2
1.45
11.9
1.32
17.2
0.75
13.4
0.28
6.7
0.03
0.8
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Lime-E
Pedal
Average Per Month
(b) All 2019 Rides by Vehicle Type
Pedal
10%
Lime-E
90%

Data and image source: 2019 Lime Bike Report (Qiu and Chang, 2020).

A3. Raw trajectory sample

trip ID	Timestamp	Latitude	Longitude
……	……	……	……
1865ec0229444baa	2019/11/21 16:50:56	42.4470900	−76.4880500
1865ec0229444baa	2019/11/21 16:50:57	42.4471200	−76.4881100
1865ec0229444baa	2019/11/21 16:50:58	42.4471300	−76.4881600
1865ec0229444baa	2019/11/21 16:50:59	42.4471400	−76.4882000
1865ec0229444baa	2019/11/21 16:51:00	42.4471600	−76.4882600
1865ec0229444baa	2019/11/21 16:51:01	42.4471900	−76.4883000
1865ec0229444baa	2019/11/21 16:52:02	42.4472200	−76.4883200
……	……	……	……

A4. Validation: Random sample of GPS raw trajectories and matched routes



A5. Shortest path algorithm

Dijkstra's algorithm is a widely used approach to finding the shortest path between any two vertices of a graph. $P = \{P_1, P_2, P_3, \cdots, P_{t-1}, P_t, \cdots\}$ denotes a set of raw GPS points of one route. After applying topology algorithm, we created a graph structure G to represent the road network. Each road segmentation is an edge of the graph. Each intersection point of the road segmentations is a node of the graph and is given a unique ID according to the logical connection of road segmentations. Q denotes the set of all nodes in G.

R'_{t-1} denotes the road segmentation on which P_{t-1} matched. R'_t denotes the road segmentation on which P_t matched. *Source* refers to the start vertex of R'_{t-1} and *Target* refers to the start vertex of R'_{t-1}. The followings are the pseudo code for Dijkstra algorithm:

```
function Dijkstra(Graph, Source, Target):
      for each vertex v in Graph:
      dist [v] ← infinity
      prev[v] ← undefined
      add v to Q
      dist[Source] ← 0
      while Q is not not empty:
      u ← vertex in Q with min dist[u]
      remove u from Q
    S ← empty sequence
    u ← Target
    if prev[u] is defined or u = Source:
         while u is defined:
      insert u at the beginning of S
      u ← prev[u]
    for each neighbor v of u still in Q:
         alt ← dist[u] + dist_between (u, v)
         if alt < dist[v]:
         dist[v] ← alt
         prev[v] ← u
  return prev[ ], dist[ ]
```

In this way, we obtain a set of the road segmentation ID between R'_{t-1} and R'_{-1}. We iterate this searching process for $P = \{P_1, P_2, P_3, \cdots, P_{t-1}, P_t, \cdots\}$ and finally reconstruct the chain of road segments for each route.

A6. Analyzing the impacts of weather conditions

Methodology

The weather data (accessed in Feb. 2024) was requested from the National Oceanic and Atmospheric Administration (NOAA) (www.ncdc.noaa.gov) for the exact spanned dates of the DBS data using the Cornell University meteorological station as location, which is the closest weather monitoring station in the study area. Because our trip records only cover a section (Nov.2019 to Mar. 2020) of the year and predominantly span the winter months, we applied the Augmented Dickey-Fuller (ADF) test to understand whether these weather data were relatively stable and stationary longitudinally. Finally, we aggregated the DBS trip data to compute the daily trip volume, using a separate OLS regression to understand the general influence of the weather conditions on the DBS trip volume.

Results

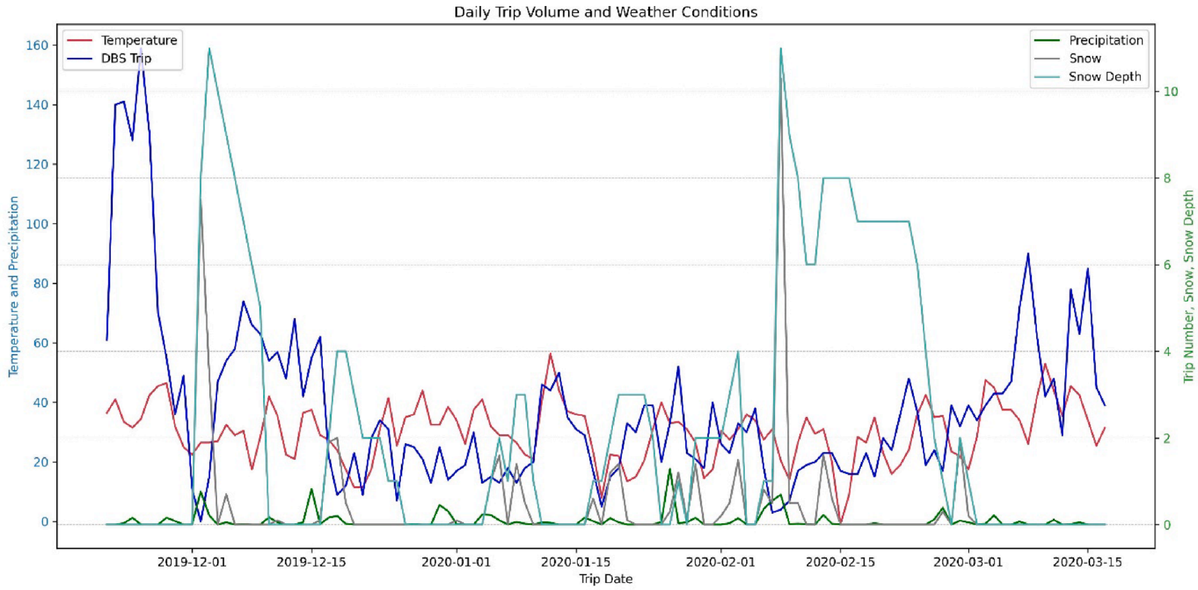
The descriptive statistics of the daily cycling volume and weather conditions are reported in the Table below. The ADF test statistics of four weather conditions variables further indicate that during this reported period, the weather was relatively stationary at the 1 % significance level. This successfully rejects the null hypothesis that a unit root is present in the time series data. These findings imply that the weather in the winter months was relatively stable and may pose a consistent impact on daily DBS travel usage.

Table A1
Descriptive statistics of all variables used in the analysis.

Dep. Variable	**Mean**	**S.D.**	**Max**	**Min**	**N**
Daily DBS cycling volume (Nov.21st, 2019 to Mar.16th, 2020)	37.542	28.677	159	0	118
Ind. Variables					
Weather Attributes (Daily)					
Ave Temperature (F)	30.118	9.808	−1.000	56.500	118
Precipitation (Inch)	0.082	0.186	1.290	0	118
Snow (Inch)	0.394	1.268	10.300	0	118
Snow Depth (Inch)	2.279	3.065	11.000	0	118

We further plotted the map (see Appendix 7) of these weather variables on par with the daily DBS trip volume; it is observed that only the average daily temperature follows a similar trend with the volume, while correlations with other variables cannot be identified visually. Therefore, an OLS regression was conducted between trip volume and weather variables to demystify the influences further. The OLS model indicates that the weather conditions can explain 16.7 % of the total variance of daily DBS volume and is further elaborated in the discussion section.

A7. Daily DBS trip volume and associated weather conditions



References

Ashqar, H.I., Elhenawy, M., Rakha, H.A., 2019. Modeling bike counts in a bike-sharing system considering the effect of weather conditions. Case Stud. Transp. Policy 7, 261–268. https://doi.org/10.1016/j.cstp.2019.02.011.

Bai, Q., Yu, Z., Ma, S., Wang, Y., Agbelie, B., 2022. Examining influencing factors of bicycle usage for dock-based public bicycle sharing system: a case study of Xi'an China. J. Clean. Prod. 362, 132332 https://doi.org/10.1016/j.jclepro.2022.132332.

Basu, R., Sevtsuk, A., 2022. How do street attributes affect willingness-to-walk? City-wide pedestrian route choice analysis using big data from Boston and San Francisco. Transp. Res. Part Policy Pract. 163, 1–19. https://doi.org/10.1016/j.tra.2022.06.007.

Bertolini, L., 2020. From "streets for traffic" to "streets for people": can street experiments transform urban mobility? Transp. Rev. 40, 734–753. https://doi.org/10.1080/01441647.2020.1761907.

Bike Share in the U.S.: 2017 [WWW Document], n.d. . Natl. Assoc. City Transp. Off. URL https://nacto.org/bike-share-statistics-2017/ (accessed 10.15.23).

Blitz, A., 2021. How does the individual perception of local conditions affect cycling? An analysis of the impact of built and non-built environment factors on cycling behaviour and attitudes in an urban setting. Travel Behav. Soc. 25, 27–40. https://doi.org/10.1016/j.tbs.2021.05.006.

Buehler, R., 2012. Determinants of bicycle commuting in the Washington, DC region: The role of bicycle parking, cyclist showers, and free car parking at work. Transp. Res. Part Transp. Environ. 17, 525–531. https://doi.org/10.1016/j.trd.2012.06.003.

Chen, J., Li, W., Zhang, H., Jiang, W., Li, W., Sui, Y., Song, X., Shibasaki, R., 2020a. Mining urban sustainable performance: GPS data-based spatio-temporal analysis on on-road braking emission. J. Clean. Prod. 270, 122489 https://doi.org/10.1016/j.jclepro.2020.122489.

Chen, Z., Van Lierop, D., Ettema, D., 2020b. Dockless bike-sharing systems: what are the implications? Transp. Rev. 40, 333–353. https://doi.org/10.1080/01441647.2019.1710306.

Chen, E., Ye, Z., 2021. Identifying the nonlinear relationship between free-floating bike sharing usage and built environment. J. Clean. Prod. 280, 124281 https://doi.org/10.1016/j.jclepro.2020.124281.

Cui, Q., Zhang, Y., Yang, G., Huang, Y., Chen, Y., 2023. Analysing gender differences in the perceived safety from street view imagery. Int. J. Appl. Earth Obs. Geoinformation 124, 103537. https://doi.org/10.1016/j.jag.2023.103537.

Dane, G., Feng, T., Luub, F., Arentze, T., 2020. Route Choice Decisions of E-bike Users: Analysis of GPS Tracking Data in the Netherlands, in: Kyriakidis, P., Hadjimitsis, D., Skarlatos, D., Mansourian, A. (Eds.), Geospatial Technologies for Local and Regional Development, Lecture Notes in Geoinformation and Cartography. Springer International Publishing, Cham, pp. 109–124. doi: 10.1007/978-3-030-14745-7_7.

Matos, F.L. de, Fernandes, J.M., Sampaio, C., Macedo, J., Coelho, M.C., Bandeira, J., 2021. Development of an information system for cycling navigation. Transp. Res. Procedia, 23rd EURO Working Group on Transportation Meeting, EWGT 2020, 16-18 September 2020, Paphos, Cyprus 52, 107–114. doi: 10.1016/j.trpro.2021.01.012.

Ding, D., Gebel, K., 2012. Built environment, physical activity, and obesity: What have we learned from reviewing the literature? Health Place, Active Living Research 18, 100–105. https://doi.org/10.1016/j.healthplace.2011.08.021.

Dong, L., Jiang, H., Li, W., Qiu, B., Wang, H., Qiu, W., 2023. Assessing impacts of objective features and subjective perceptions of street environment on running amount: A case study of Boston. Landsc. Urban Plan. 235, 104756 https://doi.org/10.1016/j.landurbplan.2023.104756.

Dubey, A., Naik, N., Parikh, D., Raskar, R., Hidalgo, C.A., 2016. Deep Learning the City: Quantifying Urban Perception at a Global Scale, in: Leibe, B., Matas, J., Sebe, N., Welling, M. (Eds.), Computer Vision – ECCV 2016, Lecture Notes in Computer Science. Springer International Publishing, Cham, pp. 196–212. doi: 10.1007/978-3-319-46448-0_12.

El-Assi, W., Salah Mahmoud, M., Nurul Habib, K., 2017. Effects of built environment and weather on bike sharing demand: a station level analysis of commercial bike sharing in Toronto. Transportation 44, 589–613. https://doi.org/10.1007/s11116-015-9669-z.

Elhorst, J.P., 2010. Applied Spatial Econometrics: Raising the Bar. Spat. Econ. Anal. 5, 9–28. https://doi.org/10.1080/17421770903541772.

Ettema, D., 2016. Runnable Cities: How Does the Running Environment Influence Perceived Attractiveness, Restorativeness, and Running Frequency? Environ. Behav. 48, 1127–1147. https://doi.org/10.1177/0013916515596364.

Ewing, R., Handy, S., 2009. Measuring the Unmeasurable: Urban Design Qualities Related to Walkability. J. Urban Des. 14, 65–84. https://doi.org/10.1080/13574800802451155.

Faghih-Imani, A., Eluru, N., 2015. Analysing bicycle-sharing system user destination choice preferences: Chicago's Divvy system. J. Transp. Geogr. 44, 53–64. https://doi.org/10.1016/j.jtrangeo.2015.03.005.

Faghih-Imani, A., Eluru, N., 2016. Incorporating the impact of spatio-temporal interactions on bicycle sharing system demand: A case study of New York CitiBike system. J. Transp. Geogr. 54, 218–227. https://doi.org/10.1016/j.jtrangeo.2016.06.008.

Faghih-Imani, A., Eluru, N., El-Geneidy, A.M., Rabbat, M., Haq, U., 2014. How land-use and urban form impact bicycle flows: evidence from the bicycle-sharing system (BIXI) in Montreal. J. Transp. Geogr. 41, 306–314. https://doi.org/10.1016/j.jtrangeo.2014.01.013.

Gao, W., Hu, X., Wang, N., 2023. Exploring spatio-temporal pattern heterogeneity of dockless bike-sharing system: Links with cycling environment. Transp. Res. Part Transp. Environ. 117, 103657 https://doi.org/10.1016/j.trd.2023.103657.

Goodman, A., Sahlqvist, S., Ogilvie, D., 2014. New Walking and Cycling Routes and Increased Physical Activity: One- and 2-Year Findings From the UK iConnect Study. Am. J. Public Health 104, e38–e46. https://doi.org/10.2105/AJPH.2014.302059.

Griffin, G., Sener, I., 2016. Planning for Bike Share Connectivity to Rail Transit. J. Public Transp. 19 https://doi.org/10.5038/2375-0901.19.2.1.

Grond, K.A., 2016. Route Choice Modeling of Cyclists in Toronto (Thesis).

Gu, T., Kim, I., Currie, G., 2019. To be or not to be dockless: Empirical analysis of dockless bikeshare development in China. Transp. Res. Part Policy Pract. 119, 122–147. https://doi.org/10.1016/j.tra.2018.11.007.

Guo, Y., He, S.Y., 2020. Built environment effects on the integration of dockless bike-sharing and the metro. Transp. Res. Part Transp. Environ. 83, 102335 https://doi.org/10.1016/j.trd.2020.102335.

Guo, Y., He, S.Y., 2021. Perceived built environment and dockless bikeshare as a feeder mode of metro. Transp. Res. Part Transp. Environ. 92, 102693 https://doi.org/10.1016/j.trd.2020.102693.

Guo, Y., Yang, L., Chen, Y., 2022. Bike Share Usage and the Built Environment: A Review. Front. Public Health 10, 848169. https://doi.org/10.3389/fpubh.2022.848169.

Hogendorf, M., Oude Groeniger, J., Noordzij, J.M., Beenackers, M.A., van Lenthe, F.J., 2020. Longitudinal effects of urban green space on walking and cycling: A fixed effects analysis. Health Place 61, 102264. https://doi.org/10.1016/j.healthplace.2019.102264.

Ito, K., Biljecki, F., 2021. Assessing bikeability with street view imagery and computer vision. Transp. Res. Part C Emerg. Technol. 132, 103371 https://doi.org/10.1016/j.trc.2021.103371.

Jacobs, J., 1961. The Death and Life of Great American Cities. Random House, New York.

Jiang, H., Dong, L., Qiu, B., 2022a. How Are Macro-Scale and Micro-Scale Built Environments Associated with Running Activity? The Application of Strava Data and Deep Learning in Inner London. ISPRS Int. J. Geo-Inf. 11, 504. https://doi.org/10.3390/ijgi11100504.

Jiang, Y., Han, Y., Liu, M., Ye, Y., 2022b. Street vitality and built environment features: A data-informed approach from fourteen Chinese cities. Sustain. Cities Soc. 79, 103724 https://doi.org/10.1016/j.scs.2022.103724.

Jin, S.T., Wang, L., Sui, D., 2023. How the built environment affects E-scooter sharing link flows: a machine learning approach. J. Transp. Geogr. 112, 103687 https://doi.org/10.1016/j.jtrangeo.2023.103687.

Kang, S., Chung, Y., Yang, B., Lee, H., Lee, J., Kim, J., 2024. User preference and willingness-to-pay for operation strategies that enhance safety and convenience of E-scooter sharing services. Transp. Policy 146, 31–41. https://doi.org/10.1016/j.tranpol.2023.11.005.

Koh, P.P., Wong, Y.D., 2013. Influence of infrastructural compatibility factors on walking and cycling route choices. J. Environ. Psychol. 36, 202–213. https://doi.org/10.1016/j.jenvp.2013.08.001.

Law, S., Sakr, F.L., Martinez, M., 2014. Measuring the Changes in Aggregate Cycling Patterns between 2003 and 2012 from a Space Syntax Perspective. Behav. Sci. 4, 278–300. https://doi.org/10.3390/bs4030278.

Lin, J.-J., Zhao, P., Takada, K., Li, S., Yai, T., Chen, C.-H., 2018. Built environment and public bike usage for metro access: A comparison of neighborhoods in Beijing, Taipei, and Tokyo. Transp. Res. Part Transp. Environ. 63, 209–221. https://doi.org/10.1016/j.trd.2018.05.007.

Liu, H.-C., Lin, J.-J., 2022. Associations of built environments with spatiotemporal patterns of shared scooter use: a comparison with shared bike use. Transp. Policy 126, 107–119. https://doi.org/10.1016/j.tranpol.2022.07.012.

Lu, Y., Yang, Y., Sun, G., Gou, Z., 2019. Associations between overhead-view and eye-level urban greenness and cycling behaviors. Cities 88, 10–18. https://doi.org/10.1016/j.cities.2019.01.003.

Lue, G., Miller, E.J., 2019. Estimating a Toronto pedestrian route choice model using smartphone GPS data. Travel Behav. Soc. 14, 34–42. https://doi.org/10.1016/j.tbs.2018.09.008.

Lynch, K., 1960. The image of the city. MIT Press, Cambridge, Mass.[u.a.

Ma, L., Dill, J., 2015. Associations between the objective and perceived built environment and bicycling for transportation. J. Transp. Health 2, 248–255. https://doi.org/10.1016/j.jth.2015.03.002.

Ma, L., Dill, J., 2017. Do people's perceptions of neighborhood bikeability match "reality"? J. Transp. Land Use 10, 291–308.

Ma, L., Dill, J., Mohr, C., 2014. The objective versus the perceived environment: what matters for bicycling? Transportation 41, 1135–1152. https://doi.org/10.1007/s11116-014-9520-y.

Ma, X., Ma, C., Wu, C., Xi, Y., Yang, R., Peng, N., Zhang, C., Ren, F., 2021. Measuring human perceptions of streetscapes to better inform urban renewal: a perspective of scene semantic parsing. Cities 110, 103086. https://doi.org/10.1016/j.cities.2020.103086.

Mandhan, S., Gregg, K., 2023. Managing the curb – Public space and use of curbside cafes during the Coronavirus pandemic. Cities 132, 104070. https://doi.org/10.1016/j.cities.2022.104070.

Martin, E.W., Shaheen, S.A., 2014. Evaluating public transit modal shift dynamics in response to bikesharing: a tale of two U.S. cities. J. Transp. Geogr. 41, 315–324. https://doi.org/10.1016/j.jtrangeo.2014.06.026.

Mehrabian, A., Russell, J.A., 1974. An approach to environmental psychology. the MIT Press.

Mertens, L., Compernolle, S., Gheysen, F., Deforche, B., Brug, J., Mackenbach, J.D., Lakerveld, J., Oppert, J.-M., Feuillet, T., Glonti, K., Bárdos, H., De Bourdeaudhuij, I., 2016. Perceived environmental correlates of cycling for transport among adults in five regions of Europe. Obes. Rev. 17, 53–61. https://doi.org/10.1111/obr.12379.

Montgomery, J., 1998. Making a city: Urbanity, vitality and urban design. J. Urban Des. 3, 93–116. https://doi.org/10.1080/13574809808724418.

Naik, N., Philipoom, J., Raskar, R., Hidalgo, C., 2014. Streetscore – Predicting the Perceived Safety of One Million Streetscapes. In: In: 2014 IEEE Conference on Computer Vision and Pattern Recognition Workshops. Presented at the 2014 IEEE Conference on Computer Vision and Pattern Recognition Workshops, pp. 793–799. https://doi.org/10.1109/CVPRW.2014.121.

Nasri, A., Zhang, L., Younes, H., 2020. Analysis of the effect of multi-level urban form on bikeshare demand: Evidence from seven large metropolitan areas in the United States. J. Transp. Land Use 13, 389–408.

Noland, R.B., Smart, M.J., Guo, Z., 2016. Bikeshare trip generation in New York City. Transp. Res. Part Policy Pract. 94, 164–181. https://doi.org/10.1016/j.tra.2016.08.030.

Noland, R.B., Iacobucci, E., Zhang, W., 2023. Public Views on the Reallocation of Street Space Due to COVID-19. J. Am. Plann. Assoc. 89, 93–106. https://doi.org/10.1080/01944363.2022.2058595.

Nosal, T., Miranda-Moreno, L.F., 2014. The effect of weather on the use of North American bicycle facilities: a multi-city analysis using automatic counts. Transp. Res. Part Policy Pract. 66, 213–225. https://doi.org/10.1016/j.tra.2014.04.012.

Ogilvie, F., Goodman, A., 2012. Inequalities in usage of a public bicycle sharing scheme: socio-demographic predictors of uptake and usage of the London (UK) cycle hire scheme. Prev. Med. 55, 40–45. https://doi.org/10.1016/j.ypmed.2012.05.002.

Panter, J., Ogilvie, D., 2015. Theorising and testing environmental pathways to behaviour change: natural experimental study of the perception and use of new infrastructure to promote walking and cycling in local communities. BMJ Open 5, e007593.

Park, K., Ewing, R., Sabouri, S., Larsen, J., 2019. Street life and the built environment in an auto-oriented US region. Cities 88, 243–251. https://doi.org/10.1016/j.cities.2018.11.005.

Porter, A.K., Kohl, H.W., Pérez, A., Reininger, B., Pettee Gabriel, K., Salvo, D., 2018. Perceived Social and Built Environment Correlates of Transportation and Recreation-Only Bicycling Among Adults. Prev. Chronic. Dis. 15, E135. https://doi.org/10.5888/pcd15.180060.

Pucher, J., Buehler, R., 2016. Safer Cycling Through Improved Infrastructure. Am. J. Public Health 106, 2089–2091. https://doi.org/10.2105/AJPH.2016.303507.

Qiu, W., Chang, H., 2020. 2019 Lime Trp Data Analysis Annual Report for Bike Walk Tompkins. Bike Walk Tompkins, Ithaca, NY. doi: 10.13140/RG.2.2.30084.45444.

Qiu, W., Chang, H., 2021. The interplay between dockless bikeshare and bus for small-size cities in the US: A case study of Ithaca. J. Transp. Geogr. 96, 103175 https://doi.org/10.1016/j.jtrangeo.2021.103175.

Qiu, W., Zhang, Z., Liu, X., Li, W., Li, X., Xu, X., Huang, X., 2022. Subjective or objective measures of street environment, which are more effective in explaining housing prices? Landsc. Urban Plan. 221, 104358 https://doi.org/10.1016/j.landurbplan.2022.104358.

Qiu, W., Li, W., Liu, X., Zhang, Z., Li, X., Huang, X., 2023. Subjective and objective measures of streetscape perceptions: Relationships with property value in Shanghai. Cities 132, 104037. https://doi.org/10.1016/j.cities.2022.104037.

Ramírez, T., Hurtubia, R., Lobel, H., Rossetti, T., 2021. Measuring heterogeneous perception of urban space with massive data and machine learning: An application to safety. Landsc. Urban Plan. 208, 104002 https://doi.org/10.1016/j.landurbplan.2020.104002.

Rey, S.J., Anselin, L., 2010. PySAL: A Python Library of Spatial Analytical Methods, in: Fischer, M.M., Getis, A. (Eds.), Handbook of Applied Spatial Analysis: Software Tools, Methods and Applications. Springer, Berlin, Heidelberg, pp. 175–193. doi: 10.1007/978-3-642-03647-7_11.

Risald, Mirino, A.E., Suyoto, 2017. Best routes selection using Dijkstra and Floyd-Warshall algorithm, in: 2017 11th International Conference on Information & Communication Technology and System (ICTS). Presented at the 2017 11th International Conference on Information & Communication Technology and System (ICTS), pp. 155–158. doi: 10.1109/ICTS.2017.8265662.

Salazar Miranda, A., Fan, Z., Duarte, F., Ratti, C., 2021. Desirable streets: Using deviations in pedestrian trajectories to measure the value of the built environment. Comput. Environ. Urban Syst. 86, 101563 https://doi.org/10.1016/j.compenvurbsys.2020.101563.

Salesses, P., Schechtner, K., Hidalgo, C.A., 2013. The Collaborative Image of The City: Mapping the Inequality of Urban Perception. PLOS ONE 8, e68400.

Schweizer, J., Bernardi, S., Rupi, F., 2016. Map-matching algorithm applied to bicycle global positioning system traces in Bologna. IET Intell. Transp. Syst. 10, 244–250. https://doi.org/10.1049/iet-its.2015.0135.

Sevtsuk, A., Basu, R., Li, X., Kalvo, R., 2021. A big data approach to understanding pedestrian route choice preferences: Evidence from San Francisco. Travel Behav. Soc. doi: 10.1016/j.tbs.2021.05.010.

Shaheen, S.A., Guzman, S., Zhang, H., 2010. Bikesharing in Europe, the Americas, and Asia: past, present, and future. Transp. Res. Rec. 2143, 159–167. https://doi.org/10.3141/2143-20.

Shen, Y., Zhang, X., Zhao, J., 2018. Understanding the usage of dockless bike sharing in Singapore. Int. J. Sustain. Transp. 12, 686–700. https://doi.org/10.1080/15568318.2018.1429696.

Song, Q., Li, W., Li, M., Qiu, W., 2022a. Social Inequalities in Neighborhood-Level Streetscape Perceptions in Shanghai: The Coherence and Divergence between the Objective and Subjective Measurements. doi: 10.2139/ssrn.4179127.

Song, Q., Li, W., Li, J., Wei, X., Qiu, W., 2023b. Disclosing the Impact of Micro-level Environmental Characteristics on Dockless Bikeshare Trip Volume: A Case Study of Ithaca, in: Goodspeed, R., Sengupta, R., Kyttä, M., Pettit, C. (Eds.), Intelligence for Future Cities, The Urban Book Series. Springer Nature Switzerland, Cham, pp. 125–147. doi: 10.1007/978-3-031-31746-0_8.

Song, Q., Li, M., van Ameijde, J., 2024. The effects of curbside cafés on bike travel behaviours: spatiotemporal evidence from Toronto, in: ACCELERATED DESIGN, Proceedings of the 29th International Conference of the Association for Computer Aided Architectural Design Research in Asia (CAADRIA) 2024. the Association for Computer-Aided Architectural Design Research in Asia (CAADRIA), Hong Kong, pp. 475–484. https://caadria2024.org/wp-content/uploads/2024/04/103-THE-EFFECTS-OF-CURBSIDE-CAFES-ON-BIKE-TRAVEL-BEHAVIOURS_-SPATIOTEMPORAL-EVIDENCE-FROM-TORONTO.pdf.

Song, Q., Dou, Z., Qiu, W., Li, W., Wang, J., van Ameijde, J., Luo, D., 2023a. The evaluation of urban spatial quality and utility trade-offs for Post-COVID working preferences: a case study of Hong Kong. Archit. Intell. 2, 1. https://doi.org/10.1007/s44223-022-00020-x.

Song, Q., Liu, Y., Qiu, W., Liu, R., Li, M., 2022b. Investigating the impact of perceived micro-level neighborhood characteristics on housing prices in Shanghai. Land 11, 2002. https://doi.org/10.3390/land11112002.

Stanislav, A., Chin, J.T., 2019. Evaluating livability and perceived values of sustainable neighborhood design: New Urbanism and original urban suburbs. Sustain. Cities Soc. 47, 101517 https://doi.org/10.1016/j.scs.2019.101517.

Su, N., Li, W., Qiu, W., 2023. Measuring the associations between eye-level urban design quality and on-street crime density around New York subway entrances. Habitat Int. 131, 102728 https://doi.org/10.1016/j.habitatint.2022.102728.

Van Dyck, D., Cerin, E., Conway, T.L., De Bourdeaudhuij, I., Owen, N., Kerr, J., Cardon, G., Frank, L.D., Saelens, B.E., Sallis, J.F., 2012. Perceived neighborhood environmental attributes associated with adults' transport-related walking and cycling: Findings from the USA, Australia and Belgium. Int. J. Behav. Nutr. Phys. Act. 9, 70. https://doi.org/10.1186/1479-5868-9-70.

Verma, D., Jana, A., Ramamritham, K., 2020. Predicting human perception of the urban environment in a spatiotemporal urban setting using locally acquired street view images and audio clips. Build. Environ. 186, 107340 https://doi.org/10.1016/j.buildenv.2020.107340.

Wang, J., Lindsey, G., 2019. Neighborhood socio-demographic characteristics and bike share member patterns of use. J. Transp. Geogr. 79, 102475 https://doi.org/10.1016/j.jtrangeo.2019.102475.

Wang, R., Liu, Y., Lu, Y., Zhang, J., Liu, P., Yao, Y., Grekousis, G., 2019a. Perceptions of built environment and health outcomes for older Chinese in Beijing: A big data approach with street view images and deep learning technique. Comput. Environ. Urban Syst. 78, 101386 https://doi.org/10.1016/j.compenvurbsys.2019.101386.

Wang, X., Liu, Y., Zhu, C., Yao, Y., Helbich, M., 2022. Associations between the streetscape built environment and walking to school among primary schoolchildren in Beijing China. J. Transp. Geogr. 99, 103303 https://doi.org/10.1016/j.jtrangeo.2022.103303.

Wang, R., Lu, Y., Zhang, J., Liu, P., Yao, Y., Liu, Y., 2019b. The relationship between visual enclosure for neighbourhood street walkability and elders' mental health in China: using street view images. J. Transp. Health 13, 90–102. https://doi.org/10.1016/j.jth.2019.02.009.

Wang, R., Lu, Y., Wu, X., Liu, Y., Yao, Y., 2020. Relationship between eye-level greenness and cycling frequency around metro stations in Shenzhen, China: a big data approach. Sustain. Cities Soc. 59, 102201 https://doi.org/10.1016/j.scs.2020.102201.

Wang, H., Noland, R.B., 2021. Bikeshare and subway ridership changes during the COVID-19 pandemic in New York City. Transp. Policy 106, 262–270. https://doi.org/10.1016/j.tranpol.2021.04.004.

Wang, Y., Qiu, W., Jiang, Q., Li, W., Ji, T., Dong, L., 2023. Drivers or Pedestrians, Whose Dynamic Perceptions Are More Effective to Explain Street Vitality? A Case Study in Guangzhou. Remote Sens. 15, 568. https://doi.org/10.3390/rs15030568.

Wei, J., Chen, Y., Liu, Z., Wang, Y., 2023. Correlation between the built environment and dockless bike-sharing trips connecting to urban metro stations. J. Transp. Land Use 16, 131–161. https://doi.org/10.5198/jtlu.2023.2262.

Wei, B., Zhu, L., 2023. Exploring the Impact of Built Environment Factors on the Relationships between Bike Sharing and Public Transportation: a Case Study of New York. ISPRS Int. J. Geo-Inf. 12, 293. https://doi.org/10.3390/ijgi12070293.

Wu, F., Li, W., Qiu, W., 2023. Examining non-linear relationship between streetscape features and propensity of walking to school in Hong Kong using machine learning techniques. J. Transp. Geogr. 113, 103698 https://doi.org/10.1016/j.jtrangeo.2023.103698.

Zhang, E., Hsu, W., Long, Y., Hawken, S., 2023. Understanding Bikeability: Insight into the Cycling-City Relationship Using Massive Dockless Bike-Sharing Records in Beijing, in: Goodspeed, R., Sengupta, R., Kyttä, M., Pettit, C. (Eds.), Intelligence for Future Cities, The Urban Book Series. Springer Nature Switzerland, Cham, pp. 109–123. doi: 10.1007/978-3-031-31746-0_7.
Zhang, F., Zhou, B., Liu, L., Liu, Y., Fung, H.H., Lin, H., Ratti, C., 2018. Measuring human perceptions of a large-scale urban region using machine learning. Landsc. Urban Plan. 180, 148–160. https://doi.org/10.1016/j.landurbplan.2018.08.020.
Zhao, Q., He, Y., Wang, Y., Li, W., Wu, L., Qiu, W., 2024. Investigating the civic emotion dynamics during the COVID-19 lockdown: Evidence from social media. Sustain. Cities Soc. 107, 105403 https://doi.org/10.1016/j.scs.2024.105403.
Zhao, P., Li, S., 2017. Bicycle-metro integration in a growing city: The determinants of cycling as a transfer mode in metro station areas in Beijing. Transp. Res. Part Policy Pract. 99, 46–60. https://doi.org/10.1016/j.tra.2017.03.003.
Zhao, H., Shi, J., Qi, X., Wang, X., Jia, J., 2017. Pyramid Scene Parsing Network. In: In: 2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR). Presented at the 2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), pp. 6230–6239. https://doi.org/10.1109/CVPR.2017.660.
Zhao, P., Yuan, D., Zhang, Y., 2022. The public bicycle as a feeder mode for metro commuters in the megacity Beijing: travel behavior, route environment, and socioeconomic factors. J. Urban Plan. Dev. 148, 04021064. https://doi.org/10.1061/(ASCE)UP.1943-5444.0000785.